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Bulk-Surface Models for Ligand-Receptor Dynamics

Bachelor Thesis in Mathematics

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Contents

1	Introduction	2
2	Preliminaries	3
2.1	Analytical background and notation	3
2.2	Carathéodory theory	5
2.3	Weak derivatives and Sobolev spaces	7
2.4	Weak time derivatives	10
2.5	Variational forms	12
3	Turing patterns	13
3.1	Stability analysis	13
3.2	Example: the Schnakenberg equations	14
3.3	Example: Schnakenberg with decay	15
4	A bulk-surface model	21
4.1	Biological motivation	21
4.2	Weak formulation	23
4.3	Existence and uniqueness	24
5	Numerical analysis	34
5.1	Modified model	34
5.1.1	Diffusion on the surface	34
5.1.2	Derivation of the two-species model	35
5.2	The Finite Element Method	36
5.2.1	Basic construction	36
5.2.2	Time-dependent PDEs	38
5.2.3	Bulk-surface equations	39
5.3	Implementation	39
5.4	Results	41
6	Bibliography	42

Chapter 1

Introduction

Chapter 2

Preliminaries

2.1 Analytical background and notation

We begin by specifying some notation and recalling and defining some general tools that play an important role throughout this thesis. We will be working with a wide variety of spaces, many of them having their own inner products and norms. First of all, we denote the Euclidean norm on $\mathbb{R}^m$ by $|\cdot|$, for all $m \in \mathbb{N}$. For a more general normed space X , where X is different from $\mathbb{R}^m$ for some m , we write $\|\cdot\|_X$ for its norm. When this norm is induced by an inner product, we denote this inner product by $(\cdot, \cdot)_X$. First we collect a few inequalities that are mainly used in the estimates of our existence and uniqueness proof.

Lemma 2.1 (Young's inequality)

For every $a, b \in \mathbb{R}$, we have

$$ab \leq \frac{1}{2} (a^2 + b^2).$$

More generally, for every $\varepsilon > 0$, we have

$$ab \leq \varepsilon a^2 + \frac{1}{4\varepsilon} b^2.$$

Proof. Let $a, b \in \mathbb{R}$ and $\varepsilon > 0$. Then

$$0 \leq \left(\sqrt{\varepsilon} a - \frac{1}{2\sqrt{\varepsilon}} b \right)^2 = \varepsilon a^2 - ab + \frac{1}{4\varepsilon} b^2.$$

This proves the second inequality. The first inequality is a special case of the second, with ε set to $\frac{1}{2}$. □

The following is a much more complex inequality, which is used very often in the study of differential equations to derive explicit estimates given an inequality that involves the unknown function.

move definition absolutely continuous here

Lemma 2.2 (Gronwall's lemma: differential form)

Let $T > 0$ and let η be a non-negative, absolutely continuous function on $[0, T]$, which satisfies

$$\eta'(t) \leq \phi(t)\eta(t) + \psi(t)$$

for almost every $t \in [0, T]$, where ϕ and ψ are non-negative, L^1 functions on $[0, T]$. Then

$$\eta(t) \leq e^{\int_0^t \phi(s) \, ds} \left[\eta(0) + \int_0^t \psi(s) \, ds \right].$$

for all $t \in [0, T]$.

This is known as the *differential form* of Gronwall's lemma. A proof can be found in [Eva10]. We also have an *integral form*, which is especially useful when dealing with non-differentiable functions.

Lemma 2.3 (Gronwall's lemma: integral form)

Let $T > 0$ and let $y : [0, T] \rightarrow \mathbb{R}$ and $a : [0, T] \rightarrow \mathbb{R}$ be nonnegative functions such that a and ay belong to $L^1([0, T])$ and

$$y(t) \leq C + \int_0^t y(s)a(s) \, ds$$

for some constant C and all $t \in [0, T]$. Then for any $t \in [0, T]$, we have

$$y(t) \leq Ce^{\int_0^t a(s) \, ds}.$$

This formulation can be found in [CV02].

Definition 2.4

Let $\Omega \subseteq \mathbb{R}^d$ be open. A Lebesgue measurable function $f : \Omega \rightarrow \mathbb{R}$ is said to be *locally integrable* if

$$\int_K |f| < \infty$$

for all compact subsets K of Ω . The space of all locally integrable functions is denoted by $L^1_{\text{loc}}(\Omega)$, where we identify two functions with each other if they agree almost everywhere in Ω .

An important fact that should be noted is that every function that is square-integrable is also locally integrable. That is, $L^2(\Omega) \subseteq L^1_{\text{loc}}(\Omega)$. For the sake of completeness, we will sometimes state definitions or results with the hypothesis that a function is locally integrable, only to use them on functions that are square-integrable.

Put this before FLCV

We now continue with some concepts that are often used to translate local properties into global results. This method plays a fundamental part in our later discussions about the formulation and analysis of weak forms and solutions to partial differential equations (PDEs).

Let X be a topological space. A function $\phi : X \rightarrow \mathbb{R}$ is said to have *compact support* if the set

$$\text{supp}(\phi) := \overline{\{x \in X \mid \phi(x) \neq 0\}}$$

is a compact subset of X . For any open set $\Omega \subseteq \mathbb{R}^d$, the space of all infinitely-times continuously differentiable functions with compact support in Ω is denoted by $C_c^\infty(\Omega)$.

An important property of these functions is that they must be 0 on the boundary of Ω . In the particular context of PDE analysis, they are often referred to as *test functions*. They are used in a wide range of applications, for instance in the field of the Calculus of Variations. A fundamental tool in this theory is the following result, hence its name.

Lemma 2.5 (Fundamental Lemma of the Calculus of Variations)

Let $\Omega \subseteq \mathbb{R}^d$ be open and $f \in L^1_{\text{loc}}(\Omega)$. If

$$\int_{\Omega} f \phi = 0$$

for all $\phi \in C_c^\infty(\Omega)$, then $f = 0$ almost everywhere in Ω .

Proof. For every $\varepsilon > 0$, define $\eta_\varepsilon \in C_c^\infty$ as the standard mollifier from [Eva10, Appendix C.5]. Then Theorem 7 from [Eva10, Appendix C.5] implies that

$$\int_{\Omega} \eta_\varepsilon(x - y) f(y) dy \rightarrow f(x) \quad \text{as } \varepsilon \rightarrow 0$$

for almost every $x \in \Omega$. But since for small enough ε the function $y \mapsto \eta_\varepsilon(x - y)$ is in $C_c^\infty(\Omega)$, this limit is 0 due to the assumption, i.e. $f(x) = 0$ for almost every $x \in \Omega$. $\square$

Multiplying a PDE with a test function and integrating by parts is sometimes referred to as *testing against* a test function $\phi \in C_c^\infty(\Omega)$. We will see more applications of this in Section 2.3.

When we are differentiating functions in C^k defined in a subset of $\mathbb{R}^d$, we sometimes use the abbreviated notation

$$D^\alpha := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}},$$

where $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ is called a *multi-index* and $|\alpha| := \alpha_1 + \dots + \alpha_d \leq k$.

2.2 Carathéodory theory

Next, we develop a weakened and more general notion of solutions to ordinary differential equations (ODEs).

Definition 2.6

Let $a < b$. A function $f : [a, b] \rightarrow \mathbb{R}$ is called *absolutely continuous* if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any finite collection of subintervals $\{(x_k, y_k)\}_{k=1}^N$ of $[a, b]$,

$$\sum_{k=1}^N (y_k - x_k) < \delta \implies \sum_{k=1}^N |f(y_k) - f(x_k)| < \varepsilon.$$

This notion of continuity is much stronger than the traditional one. There is a crucial equivalence, proved in [Nie97].

Proposition 2.7

A function $f : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous if and only if f has derivative f' almost everywhere, f' is integrable, and

$$f(x) = f(a) + \int_a^x f'(t) dt$$

for every $x \in [a, b]$.

We now have enough context to define a new class of ODE solutions.

Definition 2.8

Let n be a positive integer, I an interval, $x : I \rightarrow \mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}^n$ for some set $S \subseteq \mathbb{R} \times \mathbb{R}^n$. Then x is called a *solution* of

$$x'(t) = f(t, x(t)) \tag{2.1}$$

in the sense of Carathéodory on I if x is absolutely continuous, $(t, x(t)) \in S$ for every $t \in I$ and Equation (2.1) is satisfied for almost every $t \in I$.

Since this definition requires less from the solution, there is also a more general existence theorem, found in [CL55].

Theorem 2.9 (Carathéodory's existence theorem)

Let n be a positive integer, $a, b > 0$ and $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$. Define

$$R := \{(t, x) \in \mathbb{R} \times \mathbb{R}^n \mid |t - t_0| \leq a \text{ and } |x - x_0| \leq b\}$$

and let $f : R \rightarrow \mathbb{R}^n$ be continuous in x for every fixed t , measurable in t for every fixed x and bounded by some time-dependent integrable function m ,

$$|f(t, x)| \leq m(t)$$

for all $(t, x) \in R$. Then the differential equation

$$x'(t) = f(t, x(t))$$

has a solution in the sense of Carathéodory on the interval $|t - t_0| \leq \beta$ for some $\beta > 0$, satisfying $x(0) = x_0$.

Theorem 2.10 (Nagumo's invariance theorem)

Let $M \subseteq \mathbb{R}^n$ be closed and $J = [a, b]$ some real interval. Assume that $f : D \subseteq J \times M \rightarrow \mathbb{R}^n$ satisfies the tangent condition

$$(\nu(x), f(t, x)) \leq 0 \quad \text{for } t \in [\xi, \xi + a] \text{ and } x \in \partial M,$$

where $\nu(x)$ is the outward normal vector to M at x , as well as a condition of Lipschitz type

$$(x_1 - x_2, f(t, x_1) - f(t, x_2)) \leq L \|x_1 - x_2\|^2.$$

Then M is flow-invariant with respect to the differential equation $x' = f(t, x)$.

reference

2.3 Weak derivatives and Sobolev spaces

Before we talk about weak solutions to PDEs, we need to define spaces of functions that can form potential candidates for solutions. We derive this space by weakening the conditions of differentiable functions.

Consider an open and bounded domain $\Omega \subseteq \mathbb{R}^d$. Then for any continuously differentiable function $u \in C^1(\Omega)$ and test function $\phi \in C_c^\infty(\Omega)$, we have

$$\int_{\Omega} u_{x_i} \phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, dS - \int_{\Omega} u \phi_{x_i} = - \int_{\Omega} u \phi_{x_i}, \quad (2.2)$$

since ϕ is 0 on the boundary. Consequentially, for any function $u \in C^k(\Omega)$ and multi-index $\alpha \in \mathbb{N}^d$ with $|\alpha| \leq k$, we have

$$\int_{\Omega} D^\alpha u \phi = (-1)^{|\alpha|} \int_{\Omega} u D^\alpha \phi, \quad (2.3)$$

using the fact that $D^\beta \phi$ has compact support as well, for every $\beta \in \mathbb{N}^d$ with $|\beta| \leq k$. Note that for this last expression to make sense, the function $D^\alpha u$ does not necessarily have to be continuous. In fact, we can completely remove the hypothesis that u is k times continuously differentiable, and define the *weak derivative* of u as a function that satisfies the same integral Equation (2.3) for all test functions $\phi \in C_c^\infty(\Omega)$.

Definition 2.11

Let $u, v \in L^1_{\text{loc}}(\Omega)$ and $\alpha \in \mathbb{N}^d$. We say that v is the α -th weak derivative of u if

$$\int_{\Omega} v \phi = (-1)^{|\alpha|} \int_{\Omega} u D^{\alpha} \phi \quad (2.4)$$

for every test function $\phi \in C_c^{\infty}(\Omega)$.

We often use the same notation for weak derivatives as we do for regular derivatives. When $D^{\alpha}u$ is used, we should carefully think about which derivative is meant here. This notion extends to any differential operator that can classically be written in terms of partial derivatives, like the gradient ∇ or Laplacian $\Delta := \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_d^2}$. To be able to talk about *the* α -th weak derivative, we show that weak derivatives are unique.

Proposition 2.12

If the α -th weak derivative of u exists, it is unique up to a set of measure zero.

The proof of this proposition is a good example of how the Fundamental Lemma of the Calculus of Variations can be used.

Proof. Suppose that v and w are two α -th weak derivatives of u . Then for every $\phi \in C_c^{\infty}(\Omega)$,

$$\int_{\Omega} v \phi = (-1)^{|\alpha|} \int_{\Omega} u D^{\alpha} \phi = \int_{\Omega} w \phi,$$

which gives

$$\int_{\Omega} (v - w) \phi = 0$$

for every $\phi \in C_c^{\infty}(\Omega)$. We can now use the Lemma 2.5 to see that $v - w = 0$ almost everywhere, which means that $v = w$ almost everywhere, completing the proof. $\square$

With this new notion of weak differentiability, we can define new function spaces. These spaces are fundamental in the analysis of PDEs and will be used a lot in this thesis.

Definition 2.13

For $k \in \mathbb{N}$ and $p \in [1, \infty)$, the Sobolev space of order k , denoted $W^{k,p}(\Omega)$, is the space of functions for which the α -th weak derivative exists and is in $L^p(\Omega)$, for all $|\alpha| \leq k$:

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \mid D^{\alpha}u \in L^p(\Omega), |\alpha| \leq k\}.$$

It is equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}.$$

The space $W^{k,p}(\Omega)$ complete, and thus forms a Banach space.

Recall that $L^p(\Omega)$ is a Banach space for every $p \in [1, \infty]$ and additionally forms a Hilbert space for $p = 2$. This holds for Sobolev spaces as well. We write $H^k(\Omega) := W^{k,2}(\Omega)$ and equip it with the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha| \leq k} (D^\alpha u, D^\alpha v)_{L^2(\Omega)},$$

forming a Hilbert space. We write $H^k(\Omega)^*$ for its dual space and $\langle u, v \rangle$ for the pairing between $H^k(\Omega)^*$ and $H^k(\Omega)$, i.e.

$$\langle u, v \rangle := u(v)$$

for all $u \in H^k(\Omega)^*$ and $v \in H^k(\Omega)$.

In this thesis, we will be studying a bulk-surface reaction-diffusion model. This means that a function v defined on Ω will be reacting with some functions defined on a subset of the boundary $\partial\Omega$. These reactions will depend on the values of v on this boundary, but by definition, $H^1(\Omega)$ is a subset of $L^2(\Omega)$, so the functions that make up this Sobolev space are only defined up to a set of measure zero. The problem lies in the fact that the boundary is a set of measure zero, so we can generally not say anything about the values that are attained on it yet. We can, however, consider the case where we have a continuously differentiable function $u \in C^1(\overline{\Omega})$, for which the values on $\partial\Omega$ are certainly defined. We can also see u as a function in $H^1(\Omega) \cap C(\overline{\Omega})$, and define $v|_{\partial\Omega} := u|_{\partial\Omega}$ for every $v \in H^1(\Omega) \cap C(\overline{\Omega})$ that is equal to u almost everywhere in Ω . To further develop this idea, we first state the following property, proved in [Eva10].

Proposition 2.14 (Approximation by smooth functions)

If the boundary $\partial\Omega$ is C^1 and $1 \leq p < \infty$, the set of all smooth functions $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$.

More precisely, for any function $u \in W^{k,p}(\Omega)$, there is a sequence of smooth functions defined on the closure of Ω , $(u_n)_{n \in \mathbb{N}} \subseteq C^\infty(\overline{\Omega})$, such that

$$\|u - u_n\|_{W^{k,p}(\Omega)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This allows us to extend our knowledge of functions defined on the boundary to functions in $W^{k,p}(\Omega)$ in general.

Theorem 2.15 (Trace Theorem)

If the boundary $\partial\Omega$ is C^1 and $1 \leq p < \infty$, there is a bounded linear operator $\gamma : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega)$ such that

- (i) $\gamma u = u|_{\partial\Omega}$ for any $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$ and
- (ii) $\|\gamma u\|_{L^p(\partial\Omega)} \leq C_\gamma \|u\|_{W^{1,p}(\Omega)}$,

where the constant $C_\gamma > 0$ only depends on p and Ω .

A full proof of this theorem can be found in [Eva10], but we will describe the general outline here.

Proof. As we previously discussed, we can start by defining $\gamma u := u|_{\partial\Omega}$ for any $u \in C^1(\overline{\Omega})$. The first step in extending this definition is to use that the boundary is Lipschitz to locally flatten it and prove Item (ii) for $u \in C^1(\overline{\Omega})$ in a neighbourhood of some point $x \in \partial\Omega$. We can then use the fact that Ω is bounded to see that $\partial\Omega$ is compact, which allows us to get a finite number of points $x_1, \dots, x_n \in \partial\Omega$ with corresponding constants $C_1, \dots, C_n$, their neighbourhoods covering $\partial\Omega$. Combining the results for each x_i , we can conclude Item (ii) for the entirety of $\partial\Omega$.

For the general case of $u \in W^{1,p}(\Omega)$, we can use Proposition 2.14 to get an approximation for u of smooth functions $(u_n)_{n \in \mathbb{N}} \subseteq C^\infty(\overline{\Omega})$. The first part of the proof then implies that

$$\|\gamma u_n - \gamma u_m\|_{L^2(\partial\Omega)} = \|\gamma(u_n - u_m)\|_{L^2(\partial\Omega)} \leq C \|u_n - u_m\|_{W^{k,p}(\Omega)},$$

which means that $(\gamma u_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $L^2(\partial\Omega)$. We have used that γ is linear on $C^1(\Omega)$, since it simply acts as a restriction there. Using the fact that $L^2(\partial\Omega)$ is a Hilbert space, we can define

$$\gamma u := \lim_{n \rightarrow \infty} \gamma u_n = \lim_{n \rightarrow \infty} u_n|_{\partial\Omega},$$

which satisfies the same bound.

The proof is completed by noting that if $u \in W^{1,p}(\Omega) \cap C(\overline{\Omega})$, then there are smooth functions $(u_n)_{n \in \mathbb{N}} \subseteq C^\infty(\overline{\Omega})$ converging to u uniformly on $\overline{\Omega}$, so $\gamma u = u|_{\partial\Omega}$. $\square$

Definition 2.16

This operator $\gamma : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega)$ is referred to as the *trace operator*. Moreover, γu is called the *trace* of u , and is sometimes denoted by $u|_{\partial\Omega}$ or just u .

2.4 Weak time derivatives

To incorporate time into these new notions of weak differentiability, we first develop the concept of L^p -spaces of functions that are not necessarily real- or complex-valued. Instead, let X be any Banach space and $T > 0$. A function $f : [0, T] \rightarrow X$ is then called *strongly measurable* if there is a sequence of simple functions $(f_n)_{n \in \mathbb{N}}$ that converges pointwise to f at almost every $t \in [0, T]$.

Now for $p \in [1, \infty]$, the space $L^p(0, T; X)$ consists of all strongly measurable functions u such that $\|u\|_{L^p(0, T; X)} < \infty$, where

$$\|u\|_{L^p(0, T; X)} := \begin{cases} \left(\int_0^T \|u(t)\|_X^p dt \right)^{1/p} & \text{if } p \in [1, \infty), \\ \text{ess sup}_{t \in [0, T]} \|u(t)\|_X & \text{if } p = \infty. \end{cases}$$

These spaces are called *Bochner spaces*, and just as with traditional L^p -spaces, we can use them to define weak derivatives in time.

Definition 2.17

We say that $v \in L^1(0, T; X)$ is the *weak time derivative* of $u \in L^1(0, T; X)$, written $v = u'$ or $v = u_t$, when

$$\int_0^T u(t) \phi'(t) dt = - \int_0^T v(t) \phi(t) dt$$

for all $\phi \in C_c^\infty(0, T)$.

Lastly, $C([0, T]; X)$ denotes the space of all continuous functions $u : [0, T] \rightarrow X$ with

$$\|u\|_{C([0, T]; X)} := \max_{t \in [0, T]} \|u(t)\|_X < \infty.$$

define time
dependent
Sobolev
spaces

We can take X to be, for instance, $L^2(\Omega)$. Then formally, $C([0, T]; L^2(\Omega))$ consists of continuous functions u from the interval time $[0, T]$ to $L^2(\Omega)$. For a more intuitive interpretation, you might feel inclined to interpret them as the corresponding functions $\tilde{u}$ from $[0, T] \times \Omega$ to $\mathbb{R}$, given by $\tilde{u}(t, x) = u(t)(x)$. We do need to be careful when we use this interpretation, though, as the notation might make it seem like these functions $\tilde{u}$ are continuous in time, while this is generally not the case. Continuity in the L^2 -norm gives that small changes in time produce small changes in overall spatial profile, but not necessarily at an individual point $x \in \Omega$.

There is an important connection between these two new spaces, which will be very useful in our later arguments for the existence and uniqueness of solutions to our model. It is a special case of a result known as the Lions-Magenes lemma, and was originally discussed in [LM72]. A more modern version is presented in [Eva10], though this is specified to a case assuming the function u vanishes on the boundary, something which is not needed for the proof.

Theorem 2.18 (Lions-Magenes)

Suppose that $u \in L^2(0, T; H^1(\Omega))$ with $u_t \in L^2(0, T; H^1(\Omega)^*)$. Then

- (i) $u \in C([0, T]; L^2(\Omega))$ after possibly being redefined on a set of measure zero,
- (ii) the mapping $t \mapsto \|u(t)\|_{L^2(\Omega)}^2$ is absolutely continuous with derivative

$$\frac{d}{dt} \|u(t)\|_{L^2(\Omega)}^2 = 2 (u_t(t), u(t))_{L^2(\Omega)}$$

for almost every $t \in [0, T]$, and

(iii) we have the estimate

$$\|u(t)\|_{C([0,T],L^2(\Omega))} \leq C \left(\|u\|_{L^2(0,T;H^1(\Omega))} + \|u_t\|_{L^2(0,T;H^1(\Omega)^*)} \right),$$

the constant C only depending on T .

2.5 Variational forms

Using the notion of weak derivatives, we can weaken the requirements for solutions to a PDE. Consider, for instance, the heat equation in $(0, T) \times \Omega$, where $\Omega \subseteq \mathbb{R}^m$ is open and bounded with a smooth boundary, with Neumann boundary conditions,

$$\begin{aligned} u_t &= \Delta u + f && \text{in } (0, T) \times \Omega, \\ \frac{\partial u}{\partial \nu} &= g && \text{on } [0, T] \times \partial\Omega, \end{aligned} \tag{2.5}$$

where f and g are continuous functions on $(0, T) \times \Omega$ and $[0, T] \times \partial\Omega$ respectively. For a function u to be a classical solution, we require u to be of class $C^{1,2}((0, T) \times \Omega) \cap C^0([0, T] \times \bar{\Omega})$, i.e. continuously differentiable with respect to t , twice continuously differentiable with respect to x and continuously extendable to $[0, T] \times \bar{\Omega}$. We can multiply the heat equation by a smooth function $\phi \in C^\infty(\Omega)$, integrate over Ω and integrate by parts to derive

smooth, not
test function
right?

$$\int_{\Omega} u_t \phi = \int_{\Omega} \Delta u \phi + \int_{\Omega} f \phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, dS - \int_{\Omega} \nabla u \cdot \nabla \phi + \int_{\Omega} f \phi = \int_{\partial\Omega} g \phi \, dS - \int_{\Omega} \nabla u \cdot \nabla \phi + \int_{\Omega} f \phi.$$

We can also write this as

$$(u_t, \phi)_{L^2(\Omega)} + (\nabla u, \nabla \phi)_{L^2(\Omega; \mathbb{R}^d)} = (g, \phi)_{L^2(\Gamma)} + (f, \phi)_{L^2(\Omega)}, \tag{2.6}$$

and by Proposition 2.14, we can show that this should also hold for every $\phi \in H^1(\Omega)$. Now note that for Equation (2.6) to be well-defined, we don't need u to be twice continuously differentiable anymore. In fact, we don't even need ∇u to be continuous. Instead, square-integrability of both ∇u and u_t is sufficient. This means we can substantially weaken the requirements for u , and instead ask if there exists a function $u \in H^1(0, T; H^1(\Omega))$ that satisfies Equation (2.6) for every $\phi \in H^1(\Omega)$ and almost every $t \in [0, T]$.

Chapter 3

Turing patterns

Morphogenesis is the biological process in the development of organisms that is concerned with organising the cells and tissue into shapes and patterns. In [Tur52], Alan Turing discussed a class of mathematical systems modelling chemical substances reacting together and diffusing through tissue, and suggested that these dynamics could be sufficient to explain the main mechanisms of morphogenesis. The development of these patterns, referred to as *Turing patterns* [Mur03], has since been used in a wide range of applications in the biological field [Mur03, EOP⁺12] as well as other fields of science [ZPVB21]. We want to investigate whether this phenomenon can be used to account for the heterogeneous distribution of receptors on the cell membrane. In this chapter, we will briefly discuss the analytical details of this pattern formation and provide two examples of pattern forming systems that are relevant to our models discussed in Chapters 4 and 5.

3.1 Stability analysis

We describe the outline of the derivation of necessary conditions for the existence of Turing patterns given in [Mur03]. Consider a system of two chemical species, of which the concentrations are modelled by two coupled reaction-diffusion equations. Let $\Omega \subseteq \mathbb{R}^d$ be an open and bounded set with C^1 boundary and $T > 0$. Let $u(t, x)$ and $v(t, x)$ denote the concentrations of the two species at $t \in [0, T]$ and $x \in \bar{\Omega}$. A reaction-diffusion model for these concentrations u and v are of the form

$$\begin{aligned} u_t &= \Delta u + \gamma f(u, v) && \text{in } (0, T) \times \Omega, \\ v_t &= d\Delta v + \gamma g(u, v) && \text{in } (0, T) \times \Omega, \\ \frac{\partial u}{\partial \nu} &= \frac{\partial v}{\partial \nu} = 0 && \text{on } [0, T] \times \partial\Omega, \\ u(0, \cdot) &= u_0 && \text{in } \Omega, \\ v(0, \cdot) &= v_0 && \text{in } \Omega, \end{aligned} \tag{3.1}$$

where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ are the reactions, $d > 0$ is the diffusion coefficient for v , $\gamma > 0$ is a reaction scaling coefficient and $u_0, v_0 : \Omega \rightarrow \mathbb{R}$ are given initial data. Turing patterns can emerge when this type of system has a homogeneous equilibrium that is linearly stable in the absence of

diffusion, i.e. if it is a linearly stable steady state of the system

$$u_t = \gamma f(u, v), \quad v_t = \gamma g(u, v), \quad (3.2)$$

and becomes unstable due to the introduction of diffusion in Equation (3.1), referred to as a *diffusion driven instability*. The relevant steady state is the positive solution (u_0, v_0) of

$$f(u_0, v_0) = g(u_0, v_0) = 0,$$

if it exists. To first investigate the stability of the steady state in absence of diffusion, we linearise the system of Equation (3.2) around the (u_0, v_0) and compute the eigenvalues of the relevant matrix. For the steady state to be stable, we require that $\text{Re } \lambda < 0$ for every eigenvalue $\lambda \in \mathbb{C}$.

To study the instability of the steady state with respect to the system of Equation (3.1), the system is linearised and decomposed into *Fourier modes* w_k with *wave number* k ¹. We assume that the solution can be written as

$$\begin{pmatrix} u(t, x) - u_0 \\ v(t, x) - v_0 \end{pmatrix} = w(t, x) = \sum_k c_k e^{\lambda(k)t} w_k(x)$$

for coefficient $c_k \in \mathbb{R}$ and growth rate $\lambda(k) \in \mathbb{C}$ for every wave number k . If the growth rate $\lambda(k)$ has positive real part for some wave number k , the corresponding Fourier mode grows exponentially in time, causing the steady state to become unstable. Exactly which Fourier modes cause the instability depends on γ and influences the spatial scale of the patterns.

These considerations lead to the following set of necessary conditions for the reactions f and g ,

$$\begin{aligned} f_u + g_v &< 0, & f_u g_v - f_v g_u &> 0, \\ df_u + g_v &> 0, & (df_u + g_v)^2 - 4d(f_u g_v - f_v g_u) &> 0, \end{aligned} \quad (3.3)$$

where every derivative is evaluated at the steady state (u_0, v_0) . In particular, these conditions require f_u and g_v to be of opposite sign, and $d > 1$. Chemically, this is interpreted as one of the two species diffusing faster than the other. Lastly, the reaction scaling coefficient γ should be set such that the spatial scale of the patterns is comparable to the size of the spatial domain Ω .

3.2 Example: the Schnakenberg equations

The Schnakenberg equations, originally discussed in [Sch79], is an example of pattern forming reactions that will be relevant to our models of Chapters 4 and 5. They are given by

$$f(u, v) = a - u + u^2 v \quad \text{and} \quad g(u, v) = b - u^2 v, \quad (3.4)$$

where a and b are positive real numbers representing constant production of both chemical species. In [Mur03], a detailed analysis is given into the specific conditions for these reactions to form Turing patterns in one-dimensional domains. As a result, we get conditions for the parameters in the model that define a subset of the parameter space within which Turing instabilities take place, referred to as the *Turing space*. For fixed d , this subset is given by the area between the two curves

¹These Fourier nodes and wave numbers are the solutions of the problem $\Delta w_k + k^2 w_k = 0$. They depend on the shape of the spatial domain Ω as well as the boundary conditions of Equation (3.1). For more detail, see [Mur03].

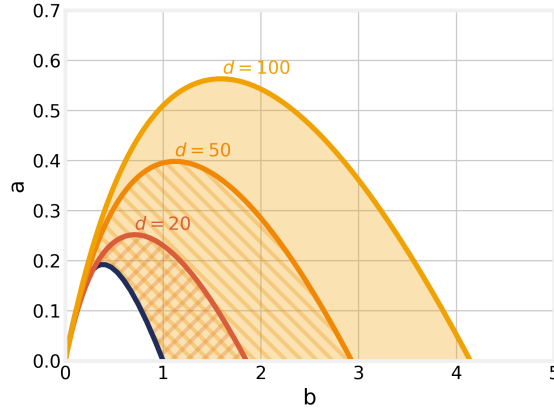


Figure 3.1: The Turing space for Equation (3.4) plotted for $d = 20$, $d = 50$ and $d = 100$.

$$\begin{aligned} a &> \frac{1}{2}u_0(1 - u_0^2), & b &= \frac{1}{2}u_0(1 + u_0^2), \\ a &< \frac{1}{2}u_0\left(1 - \frac{2u_0}{\sqrt{d}} - \frac{u_0^2}{d}\right), & b &= \frac{1}{2}u_0\left(1 + \frac{2u_0}{\sqrt{d}} + \frac{u_0^2}{d}\right), \end{aligned}$$

parameterised by u_0 , see Figure 3.1. If (a, b) is in between these two curves and γ is high enough, a Turing instability will take place. You can see that the Turing space grows as larger values of d are chosen. In Figures 3.2 and 3.3, a simulation of this model is shown with parameters within the Turing space. In this simulation, the domain is $\Omega = (-8, 8)^2$ and the parameters are set to $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The initial conditions are set to

$$u_0(x) = U + f(x) \quad \text{and} \quad v_0(x) = V + g(x), \quad (3.5)$$

where

$$U := a + b \quad \text{and} \quad V := \frac{b}{(a + b)^2} \quad (3.6)$$

are the corresponding uniform steady states and f and g are independent noise functions whose values are uniformly sampled from $(-0.1, 0.1)$.

3.3 Example: Schnakenberg with decay

The $-u$ term in Equation (3.4) could be interpreted as natural decay of the species represented by u . We show that the addition of decay of v does not cause the model to lose its ability to generate Turing patterns. To this end, consider the kinetics

$$f(u, v) = a - u + u^2v \quad \text{and} \quad g(u, v) = b - \delta_v v - u^2v, \quad (3.7)$$

Is this mixing the continuous problem and discretised simulation too much?

What is the standard way of showing these simulations?

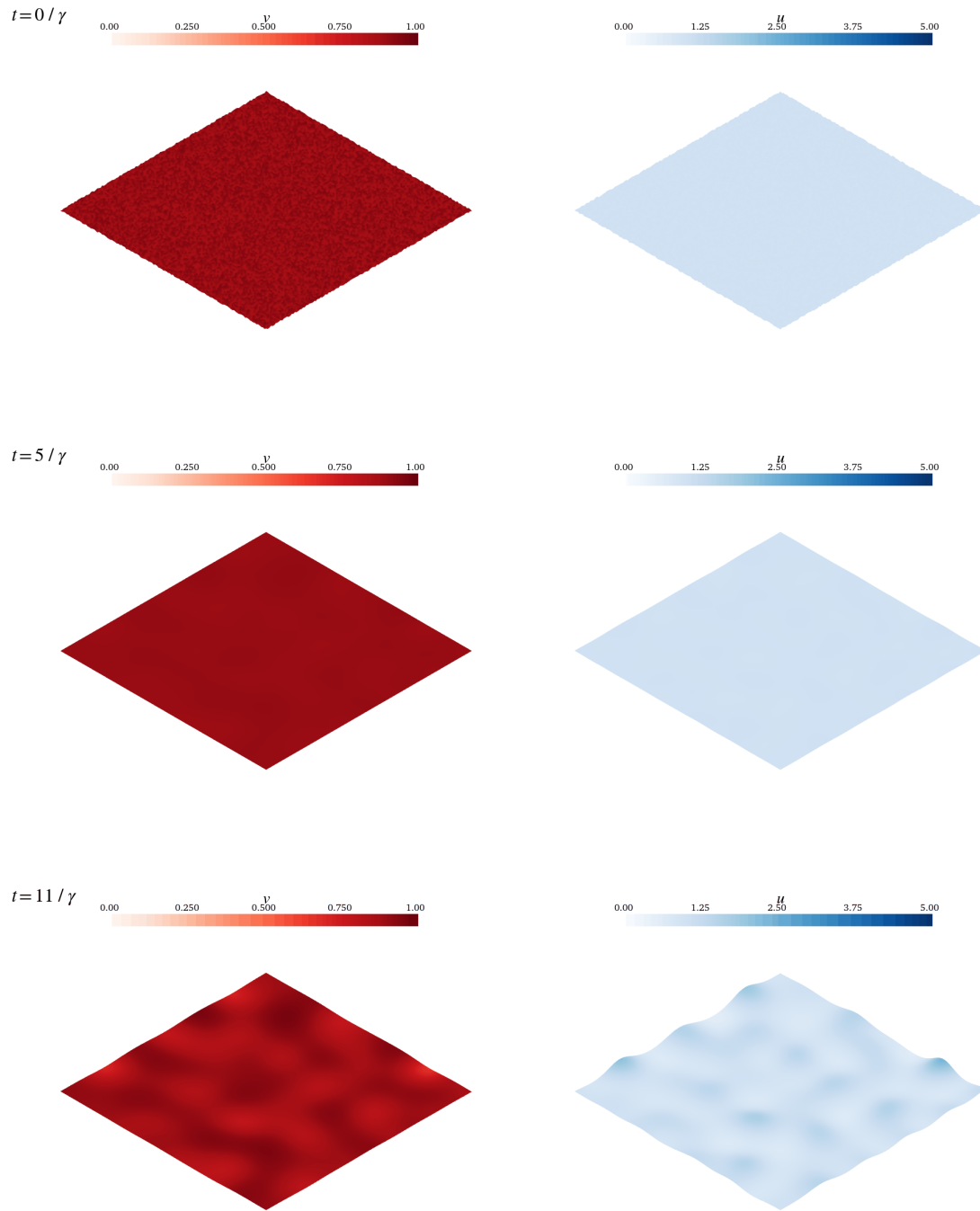


Figure 3.2: A simulation of the Schnakenberg equations on $\Omega = (-8, 8)^2$ with $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is plotted for times $t = 0$, $t = 5/\gamma$ and $t = 11/\gamma$.

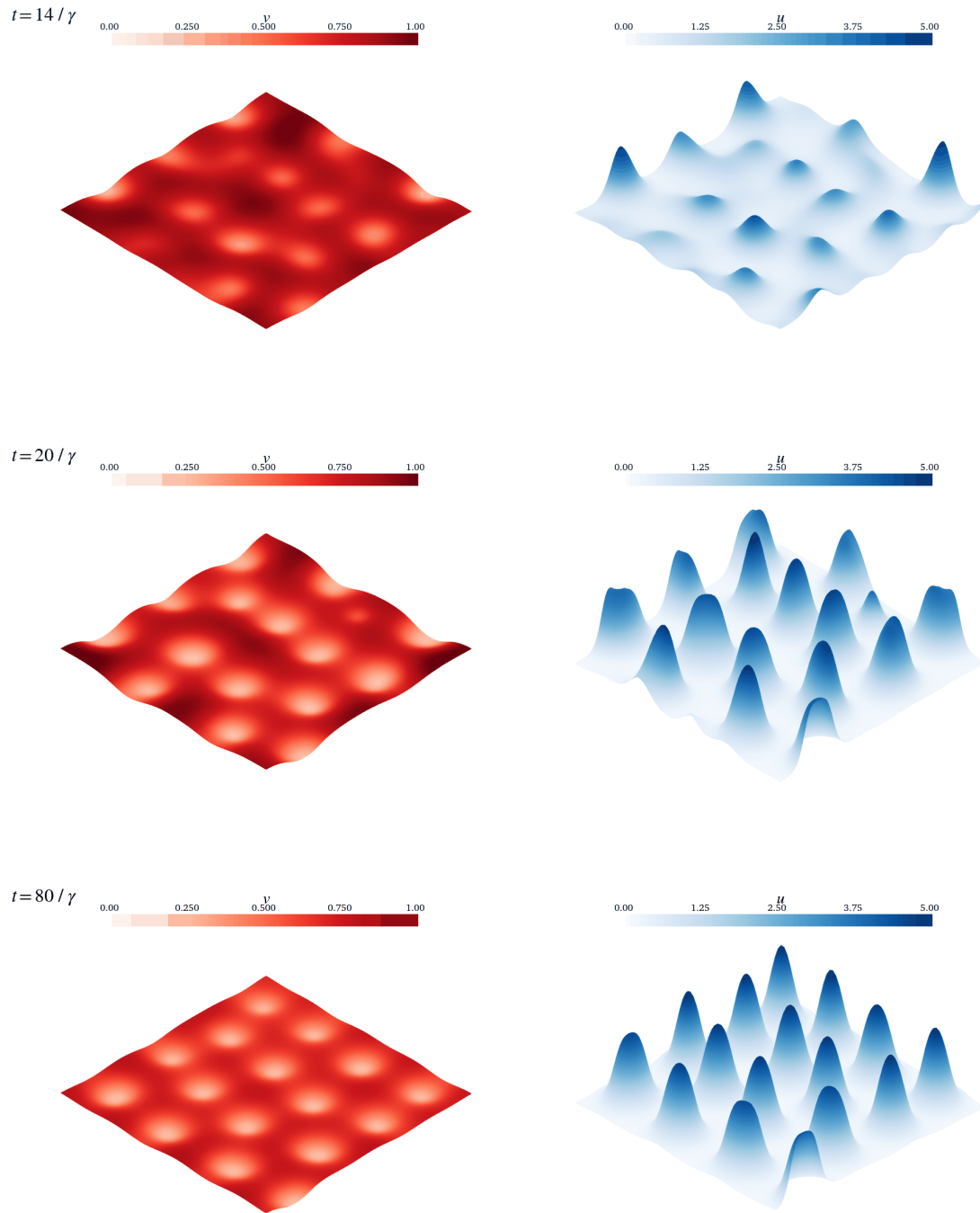


Figure 3.3: The same simulation of the Schnakenberg equations on $\Omega = (-8, 8)^2$ with $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is now plotted for times $t = 14$, $t = 20/\gamma$ and $t = 80/\gamma$.

where $\delta_v > 0$ is the decay coefficient for v . The new uniform steady states are U and V with U being a positive real root of

$$U^3 - (a + b)U^2 + \delta_v U - a\delta_v = 0,$$

if such a root exists, and

$$V := \frac{b}{U^2 + \delta_v}.$$

One can check that if $\delta_v = 0$, this returns the uniform steady states from the classical Schnakenberg equations, given in Equation (3.6). When we use these new kinetics in the model of Equation (3.1) with the new parameter δ_v set to 0.1, we see from the simulations in Figures 3.3 and 3.4 are also able to generate Turing patterns. In this simulation, the other parameters are set to the same values as before, i.e. $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The domain is $\Omega = (-8, 8)^2$ again, and the initial conditions are set to a random perturbation around the uniform steady states as in Equation (3.5). With this relatively low value of δ_v , we see that this model generates patterns that are similar to the ones generated by the classical Schnakenberg equations in both shape and scale.

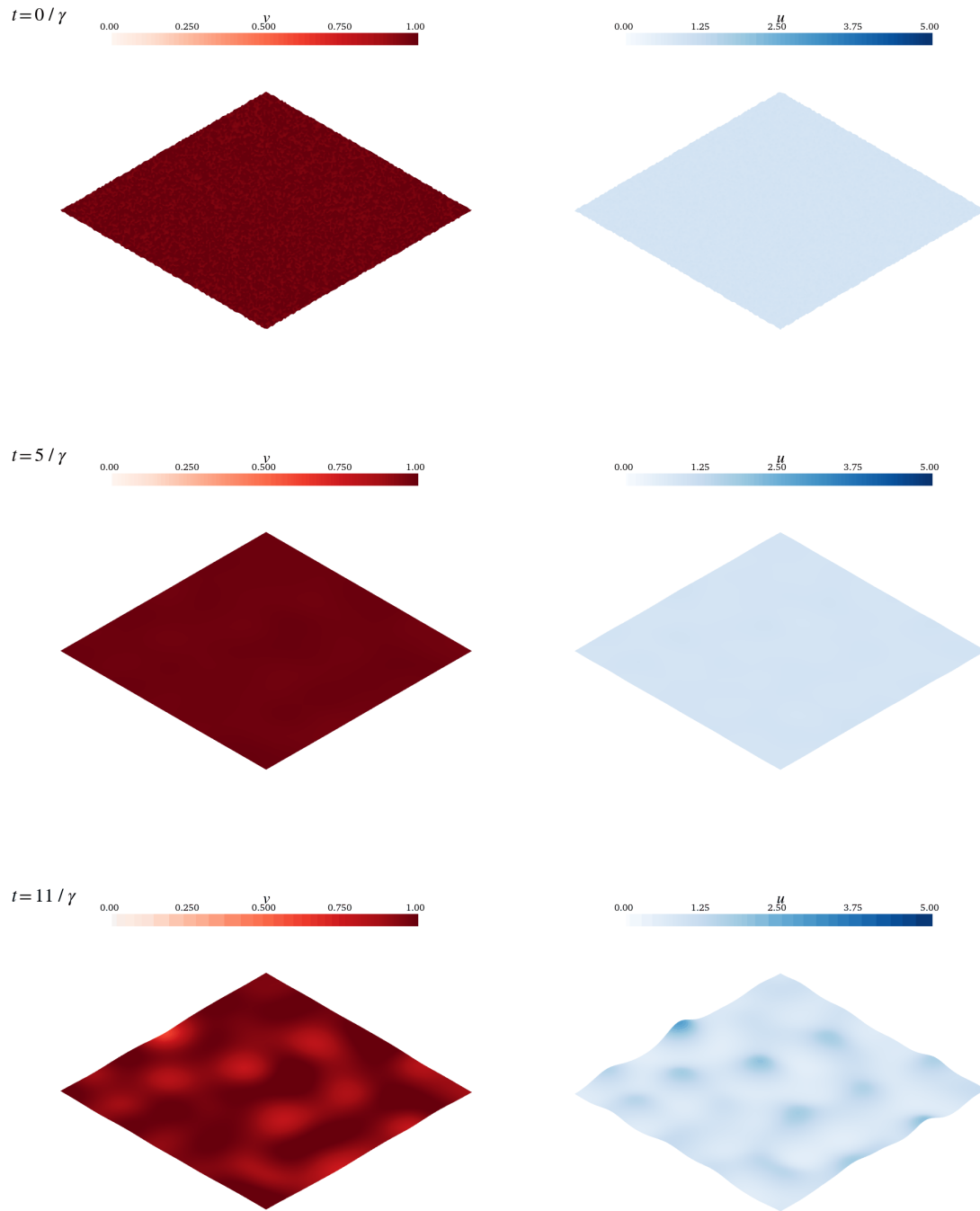


Figure 3.4: A simulation of the Schnakenberg equations with extra decay for v on $\Omega = (-8, 8)^2$ with $\delta_v = 0.1$, $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is plotted for times $t = 0$, $t = 5/\gamma$ and $t = 11/\gamma$.

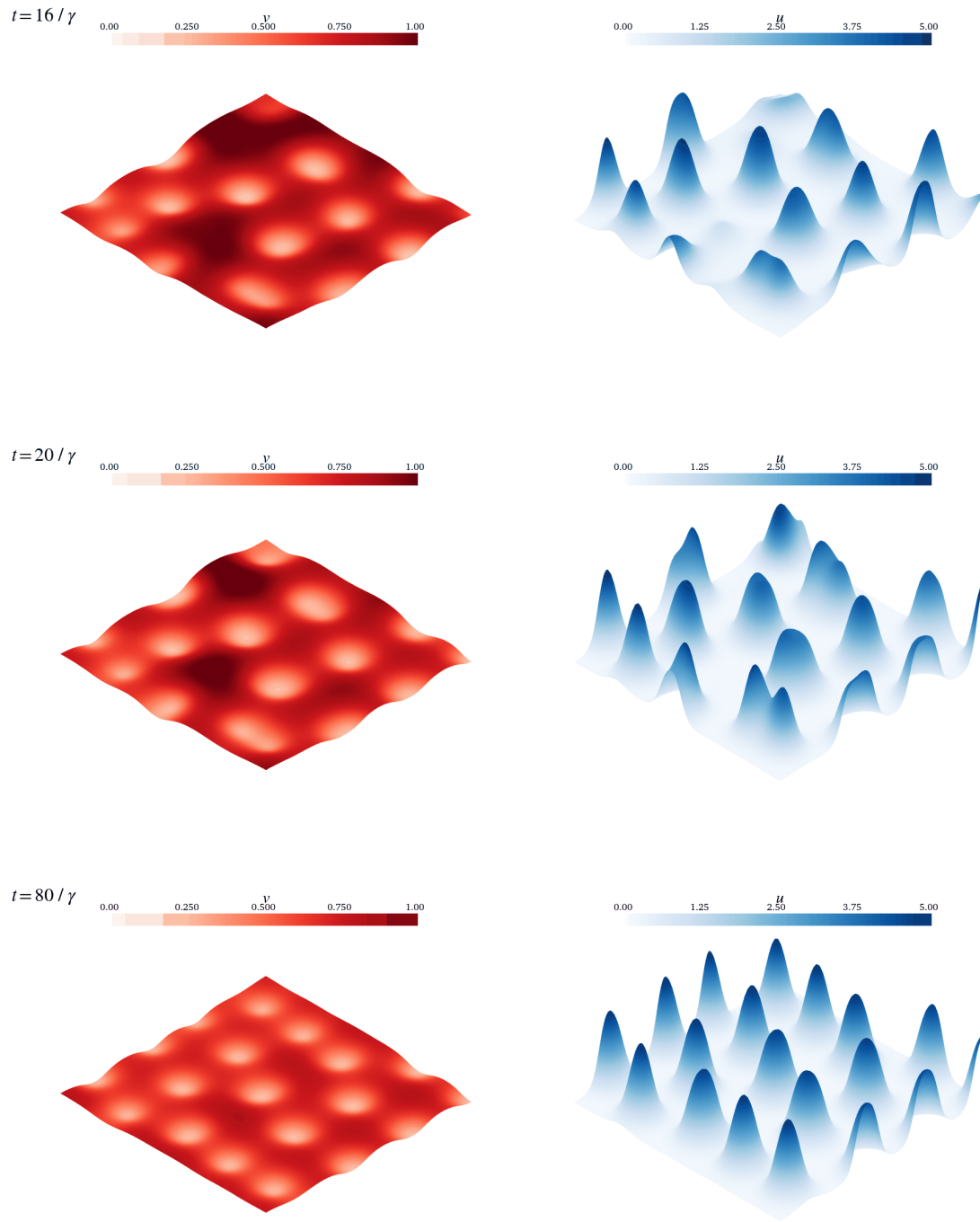


Figure 3.5: The same simulation of the Schnakenberg equations with extra decay for v on $\Omega = (-8, 8)^2$ with $\delta_v = 0.1$, $d = 50$, $a = 0.1$, $b = 0.9$ and $\gamma = 12$. The solution is now plotted for times $t = 16$, $t = 20/\gamma$ and $t = 80/\gamma$.

Chapter 4

A bulk-surface model

In this chapter, we will discuss a bulk-surface model that describes the interactions of ligands and cell receptors. We will give a proof of existence and uniqueness of weak solutions to this model, as well as discuss some of the qualitative properties of such solutions. The proof will rely heavily on the definitions and results introduced in Chapter 2, and will be based on an application of Banach's fixed-point theorem.

4.1 Biological motivation

Consider an open and bounded set $U \subseteq \mathbb{R}^d$ with C^1 boundary and $T > 0$. This can be interpreted as the microscopic part of the body we are investigating. We identify the cells in this part of the body by an open set $C \subseteq U$, also with C^1 boundary, and the extracellular space by $\Omega := U \setminus \overline{C}$. We require that the closure $\overline{C}$ of C is also completely contained in U , i.e. no cells should touch the boundary of our domain. In particular, this implies that the boundary $\partial\Omega$ of the extracellular space can be written as the disjoint union of the boundary of our region of investigation ∂U and the cell membranes ∂C , which we denote by Γ_e and Γ , respectively. We use the function $u : [0, T] \times \Gamma \rightarrow \mathbb{R}$ to represent the concentration of free receptors at a given time and location on a cell membrane. Similarly, $u_b : [0, T] \times \Gamma \rightarrow \mathbb{R}$ represents the concentration of bounded receptors, and $v : [0, T] \times \overline{\Omega} \rightarrow \mathbb{R}$ represents the concentration of ligands in $\overline{\Omega}$, see Figure 4.1.

actual biological terms

The ligands can move freely through the extracellular space. This can be modelled by the standard diffusion (or heat) equation. To simplify the setting, we assume the individual receptors will be fixed in their position on the cell membrane.

When a ligand binds to a free receptor, the concentration of free receptors and ligands decrease, and the concentration of bounded receptors at that position increases accordingly. This reaction is described by $k_{\text{on}} u^2 v$ for some $k_{\text{on}} > 0$. After some time, the bounded receptor has done its task, passing the signal through to the interior of the cell. It releases the ligand and is now available to bind to a new ligand again. The concentration of bounded receptors decrease, and the concentration of free receptors and ligands increase accordingly. The rate of this dissociation is described by $k_{\text{off}} u_b$, where $k_{\text{off}} > 0$.

Finally, two reactions that happen constantly are production and decay. To keep intercellular communication possible, cells produce ligands and receptors that enter the domain Ω through the cell membrane. We assume the production of ligands and free receptors mainly happens at constant

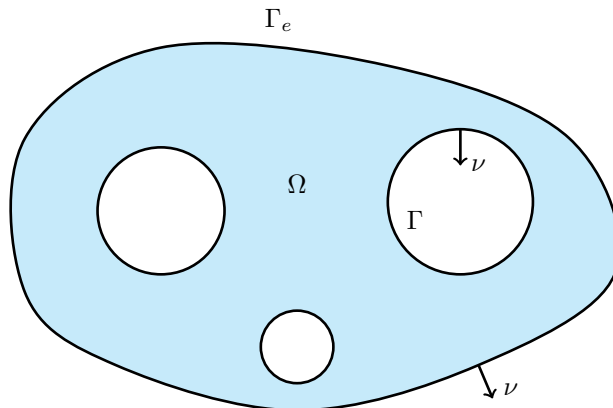


Figure 4.1: A sketch of a possible domain in the case that $d = 2$. The blue background represents Ω and the two kinds of boundaries are indicated, along with the outward unit normal vector ν .

rates $P_u > 0$ and $P_v > 0$, but is also upregulated by the concentration of bounded receptors, which is taken into account by αu_b for some upregulation coefficient $\alpha > 0$. The ligands, free receptors and complex receptors decay at constant rates δ_v , δ_u and δ_{u_b} , respectively.

We can describe the dynamics of ligands and receptors by a system of coupled differential equations. Denote the diffusion coefficient by $D_v > 0$ and the outward unit normal vector on $\partial\Omega$ by ν . Then the concentrations of receptors can then be described by the system of ODEs given by

$$\begin{aligned} u_t &= P_u + \alpha u_b - \delta_u u - 2k_{\text{on}}u^2v + 2k_{\text{off}}u_b & \text{on } (0, T) \times \Gamma, \\ u_{b,t} &= -\delta_{u_b}u_b + k_{\text{on}}u^2v - k_{\text{off}}u_b & \text{on } (0, T) \times \Gamma. \end{aligned} \quad (4.1)$$

Note that even though they are given by only two equations, this is actually a system of uncountably many ODEs. For every boundary point $x \in \Gamma$, we have one ODE for the function $t \mapsto u(t, x)$ and one for the function $t \mapsto u_b(t, x)$.

Since the dynamics of the ligands include diffusion in Ω and reaction on Γ , they can be described by the PDE and boundary conditions given by

$$\begin{aligned} v_t &= D_v \Delta v - \delta_v v & \text{in } (0, T) \times \Omega, \\ D_v \frac{\partial v}{\partial \nu} &= P_v - k_{\text{on}}u^2v + k_{\text{off}}u_b & \text{on } [0, T) \times \Gamma, \\ D_v \frac{\partial v}{\partial \nu} &= 0 & \text{on } [0, T) \times \Gamma_e. \end{aligned} \quad (4.2)$$

Additionally, we need to specify initial conditions for the concentration of ligands and receptors. That is, the solutions should satisfy

$$\begin{aligned} u(0, \cdot) &= u_0 & \text{on } \Gamma, \\ u_b(0, \cdot) &= u_{b,0} & \text{on } \Gamma, \\ v(0, \cdot) &= v_0 & \text{on } \Omega, \end{aligned} \quad (4.3)$$

for some given non-negative functions $u_0, u_{b,0} : \Gamma \rightarrow \mathbb{R}$ and $v_0 : \Omega \rightarrow \mathbb{R}$.

The remainder of this chapter will be dedicated to studying the system of Equations (4.1) to (4.3).

4.2 Weak formulation

Before we start proving the existence of solutions to this model, we must discuss what class of solutions we are considering. Specifically, we will derive the weak formulations of Equations (4.1) to (4.3) and discuss which function spaces are suitable.

We first consider the PDE, proceeding similarly to Section 2.5. To state the weak formulation, we assume $\tilde{u}$, $\tilde{u}_b$ and $\tilde{v}$ are smooth solutions to Equations (4.1) to (4.3) and note that we can also interpret them as the corresponding functions $u \in \mathcal{X}$, $u_b \in \mathcal{X}$ and $v \in \mathcal{Y}$ given by

specify $\mathcal{X}$
and $\mathcal{Y}$

$$u(t) = \tilde{u}(t, \cdot), \quad u_b(t) = \tilde{u}_b(t, \cdot) \quad \text{and} \quad v(t) = \tilde{v}(t, \cdot).$$

We multiply the PDE by a test function $\phi \in C_c^\infty(\Omega)$ and integrate over Ω , giving

$$\begin{aligned} \int_{\Omega} v_t \phi &= D_v \int_{\Omega} \Delta v \phi - \delta_v \int_{\Omega} v \phi \\ &= D_v \int_{\partial\Omega} \frac{\partial v}{\partial \nu} \phi \, dS - D_v \int_{\Omega} \nabla v \cdot \nabla \phi - \delta_v \int_{\Omega} v \phi \\ &= D_v \int_{\Gamma} \frac{\partial v}{\partial \nu} \phi \, dS + D_v \int_{\Gamma_e} \frac{\partial v}{\partial \nu} \phi \, dS - D_v \int_{\Omega} \nabla v \cdot \nabla \phi - \delta_v \int_{\Omega} v \phi \\ &= P_v \int_{\Gamma} \phi \, dS - k_{\text{on}} \int_{\Gamma} u^2 v \phi \, dS + k_{\text{off}} \int_{\Gamma} u_b \phi \, dS - D_v \int_{\Omega} \nabla v \cdot \nabla \phi - \delta_v \int_{\Omega} v \phi, \end{aligned}$$

for all $t \in (0, T)$, which we can write using inner product notation as

mention L^2
of $u^2 v$?

$$(v_t, \phi)_{L^2(\Omega)} = P_v \int_{\Gamma} \phi \, dS - k_{\text{on}} (u^2 v, \phi)_{L^2(\Gamma)} + k_{\text{off}} (u_b, \phi)_{L^2(\Gamma)} - D_v (\nabla v, \nabla \phi)_{L^2(\Omega; \mathbb{R}^d)} - \delta_v (v, \phi)_{L^2(\Omega)}. \quad (4.4)$$

Note that this also requires us to show that a solution u is essentially bounded. Otherwise, $u^2 v$ might not be of L^2 -regularity, so the term $(u^2 v, \phi)_{L^2(\Gamma)}$ would not be well-defined. Using Proposition 2.14, we can show that Equation (4.4) also holds for more general functions $\phi \in H^1(\Omega)$. Additionally, since we used integration with respect to space in this derivation, this expression only depends on time. For these reasons, a weak solution of Equation (4.2) is a function v in a suitable Sobolev space that satisfies Equation (4.4) for almost every $t \in (0, T)$ and for all $\phi \in H^1(\Omega)$.

For the ODEs, we want to use Theorem 2.9, i.e. we will be looking for solutions of the system in the sense of Carathéodory on $[0, T]$. To this end, we assume a solution $v \in \mathcal{Y}$ of the PDE exists. We fix $x \in \Gamma$ and look for an absolutely continuous function $w := (u(\cdot, x), u_b(\cdot, x))$ such that

$$w'(t) = F(t, w(t)) \quad (4.5)$$

almost everywhere, where $F : \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by

$$F(t, (u, u_b)) = \begin{pmatrix} P_u + \alpha u_b - \delta_u u - 2k_{\text{on}} u^2 \gamma v(t, x) + 2k_{\text{off}} u_b \\ -\delta_{u_b} u_b + k_{\text{on}} u^2 \gamma v(t, x) - k_{\text{off}} u_b \end{pmatrix}, \quad (4.6)$$

where γv denotes the trace of v as discussed in Section 2.3. Note that this definition of F takes into account that we assumed the existence of v . We will justify this in the next section, when we separate the problems for the ODEs and the PDE. Also note that this definition depends on the fact that x is assumed to be fixed. Thus for any $x \in \Gamma$, this defines a different ODE.

We can now combine these two formulations and start discussing the existence and uniqueness of their solutions.

define $\mathcal{X}$ and $\mathcal{Y}$

Definition 4.1

The functions $u, u_b \in \mathcal{X}$ and $v \in \mathcal{Y}$ are *weak solutions* to Equations (4.1) to (4.3) if v satisfies Equation (4.4) for almost every $t \in [0, T]$ and $\phi \in H^1(\Omega)$ and

$$w(t) = (u(t, x), u_b(t, x))$$

satisfies Equation (4.5) for almost every $t \in [0, T]$ and $x \in \Gamma$.

correct?

4.3 Existence and uniqueness

Our goal in this section is to prove the following theorem.

Theorem 4.2 (Existence and uniqueness of weak solutions)

Let the initial conditions $u_0, u_{b,0} \in L^2(\partial\Omega)$ and $v_0 \in L^2(\Omega)$ be essentially bounded and non-negative. Then there exist unique weak solutions $u, u_b \in \mathcal{X}$ and $v \in \mathcal{Y}$ to Equations (4.1) to (4.3). Moreover, these solutions are also non-negative.

As previously mentioned, the proof will be based on an application of Banach's fixed-point theorem. The proof for our model follows the same general idea as the proof of the Picard-Lindelöf theorem proving the existence and uniqueness for ODEs, but we need to adjust the arguments to handle our situation of a coupled bulk-surface system. The idea is to “freeze” the function v in the ODEs so we can use the standard existence and uniqueness theory for ODEs to get solutions u and u_b to this problem. The PDE we treat similarly, freezing the functions u and u_b . Having a solution of this PDE, we can “update” the frozen terms in the ODE. This produces an operator $\mathcal{T}$ that maps any $v \in \mathcal{Y}$ to another $\tilde{v} \in \mathcal{Y}$. If we can then show that this operator is a contraction, Banach's fixed-point theorem will guarantee the existence and uniqueness of a fixed point $v^* \in \mathcal{Y}$, which will be our solution to the PDE.

Let us first make precise what we mean by “freezing” the function v in the ODE system. Fix $v^* \in \mathcal{Y}$ and consider the modified system of ODEs given by

$$\begin{aligned} u_t &= P_u + \alpha u_b - \delta_u u - 2k_{\text{on}} u^2 v^* + 2k_{\text{off}} u_b & \text{on } (0, T) \times \Gamma, \\ u_{b,t} &= -\delta_{u_b} u_b + k_{\text{on}} u^2 v^* - k_{\text{off}} u_b & \text{on } (0, T) \times \Gamma. \end{aligned} \tag{4.7}$$

weak form?

These ODEs are now *decoupled* from the PDE, so we can investigate the existence and uniqueness of solutions to this system without simultaneously having to prove that v is a solution of the PDE.

Lemma 4.3 (Existence and uniqueness for the decoupled ODEs)

Let the initial conditions $u_0, u_{b,0} \in L^2(\Gamma)$ be essentially bounded and non-negative and fix $v^* \in \mathcal{V}$. Then there exist unique weak solutions $u, u_b \in \mathcal{X}$ to Equation (4.7) corresponding to these initial conditions. Moreover, these solutions are also non-negative and bounded by a constant depending on T .

Proof. Fix x and define w and $F : \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as in Equations (4.5) and (4.6), where v is replaced by v^* . Since the components of F are polynomial in w , the function F is certainly continuous in w . Additionally, by the regularity of γv^* , we know that F is measurable in t . This also implies that for any compact subset of $\mathbb{R} \times \mathbb{R}^2$, we can bound F by some time-dependent integrable function m . In particular, this subset can be taken to be a rectangle¹ as in the hypothesis of Theorem 2.9, so let R be such a rectangle around the initial time $t = 0$ and initial data $(u_0, u_{b,0})$. The hypotheses of Theorem 2.9 are then fully satisfied, giving us a local solution $w : [0, \beta] \rightarrow \mathbb{R}^2$ for some $\beta > 0$, with $w(0) = (u_0, u_{b,0})$.

time regularity of trace

We show that this local solution is unique. Suppose we have two such local solutions, w_1 and w_2 . Since these solutions are absolutely continuous, they are bounded on $[0, \beta]$, so we can take a compact set $K \subseteq \mathbb{R}^2$ that contains both their trajectories. For any $t \in [0, \beta]$ and $\bar{w} = (\bar{u}, \bar{u}_b), \tilde{w} = (\tilde{u}, \tilde{u}_b) \in K$, F satisfies

$$\begin{aligned} |F(t, \bar{w}) - F(t, \tilde{w})|^2 &= \left| \begin{pmatrix} -2k_{\text{on}} (\bar{u}^2 - \tilde{u}^2) v^*(t, x) + 2k_{\text{off}} (\bar{u}_b - \tilde{u}_b) + \alpha (\bar{u}_b - \tilde{u}_b) - \delta_u (\bar{u} - \tilde{u}) \\ k_{\text{on}} (\bar{u}^2 - \tilde{u}^2) v^*(t, x) - k_{\text{off}} (\bar{u}_b - \tilde{u}_b) - \delta_{u_b} (\bar{u}_b - \tilde{u}_b) \end{pmatrix} \right|^2 \\ &= \left| \begin{pmatrix} -(2k_{\text{on}} (\bar{u} + \tilde{u}) v^*(t, x) + \delta_u) (\bar{u} - \tilde{u}) & + (2k_{\text{off}} + \alpha) (\bar{u}_b - \tilde{u}_b) \\ k_{\text{on}} (\bar{u} + \tilde{u}) v^*(t, x) (\bar{u} - \tilde{u}) & - (k_{\text{off}} + \delta_{u_b}) (\bar{u}_b - \tilde{u}_b) \end{pmatrix} \right|^2 \\ &\leq 2 \left| \begin{pmatrix} -2k_{\text{on}} (\bar{u} + \tilde{u}) v^*(t, x) - \delta_u \\ k_{\text{on}} (\bar{u} + \tilde{u}) v^*(t, x) \end{pmatrix} (\bar{u} - \tilde{u}) \right|^2 + 2 \left| \begin{pmatrix} 2k_{\text{off}} + \alpha \\ k_{\text{off}} + \delta_{u_b} \end{pmatrix} (\bar{u}_b - \tilde{u}_b) \right|^2 \\ &\leq L_K(t)^2 \left((\bar{u} - \tilde{u})^2 + (\bar{u}_b - \tilde{u}_b)^2 \right) \\ &\leq L_K(t)^2 |\bar{w} - \tilde{w}|^2, \end{aligned}$$

where $L_K(t)$ is defined as

$$L_K(t) := \sqrt{2} \max \left\{ \left| \begin{pmatrix} 4k_{\text{on}} \text{diam}(K) v^*(t, x) + \delta_u \\ 2k_{\text{on}} \text{diam}(K) v^*(t, x) \end{pmatrix} \right|, \left| \begin{pmatrix} 2k_{\text{off}} + \alpha \\ k_{\text{off}} + \delta_{u_b} \end{pmatrix} \right| \right\},$$

which is of L^2 -regularity in t due to the essential boundedness of v^* . Now for the difference of the two solutions we obtain

¹On first sight, it can seem strange to call this type of set a rectangle instead of a cylinder. This name comes from the fact that in our main source [CL55], the theorem is proved for the case of $n = 1$.

$$\begin{aligned}
|w_1(t) - w_2(t)| &= \left| (u_0, u_{b,0}) + \int_0^t F(t, w_1(t)) dt - (u_0, u_{b,0}) - \int_0^t F(t, w_2(t)) dt \right| \\
&\leq \int_0^t |F(t, w_1(t)) - F(t, w_2(t))| dt \\
&\leq \int_0^t L_K(t) |w_1(t) - w_2(t)| dt
\end{aligned}$$

for every $t \in [0, \beta]$. Using Gronwall's lemma and the fact that $w_1(0) = w_2(0)$, we see that this implies that $|w_1(t) - w_2(t)| = 0$ for all $t \in [0, \beta]$, so $w_1 = w_2$.

To show that the components u and u_b of this solution w are non-negative, we use Theorem 2.10.

We now let x vary and show that u and u_b have L^2 regularity in both space and time.

We can use the non-negativity of u and u_b to show that these functions are bounded. Define $r := u + 2u_b$ and note that

$$r_t = u_t + 2u_{b,t} = P_u - \delta_u u + (\alpha - 2\delta_{u_b})u_b \leq P_u + mr,$$

where $m := \max(\delta_u, \frac{1}{2}(\alpha - 2\delta_{u_b}))$. Then Gronwall's lemma gives

$$r(t) \leq e^{mt} (r(0) + tP_u).$$

Now the non-negativity of both u and u_b and the essential boundedness of u_0 and $u_{b,0}$ allows us to bound the solutions independently by

$$\begin{aligned}
u &\leq e^{mT} (M + P_u) =: M_u(T) \quad \text{and} \\
u_b &\leq \frac{1}{2} M_u(T) =: M_{u_b}(T),
\end{aligned}$$

where $M = \text{ess sup}(u_0 + 2u_{b,0})$, for almost every $x \in \Gamma$ and $t \in [0, T]$. $\square$

We can use a similar approach for the PDE in Equation (4.2). Fix $(u^*, u_b^*) \in \mathcal{X}^2$ and consider the modified weak formulation given by

$$\begin{aligned}
(v_t, \phi)_{L^2(\Omega)} &= P_v \int_{\Gamma} \phi dS - k_{\text{on}} \left((u^*)^2 v, \phi \right)_{L^2(\Gamma)} + k_{\text{off}} (u_b^*, \phi)_{L^2(\Gamma)} \\
&\quad - D_v (\nabla v, \nabla \phi)_{L^2(\Omega; \mathbb{R}^d)} - \delta_v (v, \phi)_{L^2(\Omega)}.
\end{aligned} \tag{4.8}$$

This PDE is now decoupled from the ODEs, and we can prove the existence and uniqueness independently from the ODE system.

Lemma 4.4 (Existence and uniqueness for the decoupled PDE)

Let the initial condition $v_0 \in L^2(\Omega)$ be essentially bounded and non-negative and fix $(u^*, u_b^*) \in \mathcal{X}^2$. Then there exists a unique weak solution $v \in \mathcal{Y}$ to Equation (4.8) satisfying these initial conditions. Moreover, this solution is also non-negative and bounded by a constant depending on T .

gronwall application

Lipschitz condition?

regularity

Proof. Now let v be such a local solution. To show that v is non-negative, we estimate the negative part of v . Let v_+ and v_- be defined as $\max(0, v)$ and $-\min(0, v)$ respectively, such that $v = v_+ - v_-$. Then as discussed in [Eva10], we have $v_+, v_- \in L^2(0, T; H^1(\Omega))$ and $(v_+)_t, (v_-)_t \in L^2(0, T; H^1(\Omega)^*)$. Furthermore, $v_t v_- = -(v_-)_t v_-$, as well as $\nabla v \cdot \nabla v_- = -|\nabla v_-|^2$, so for all almost every t we have

Existence and Uniqueness

reference?

$$\frac{d}{dt} \frac{1}{2} \|v_-\|_{L^2(\Omega)}^2 = \frac{d}{dt} \frac{1}{2} \int_{\Omega} v_-^2 = \int_{\Omega} (v_-)_t v_- = - \int_{\Omega} v_t v_-$$

due to Theorem 2.18. We use v_- as a test function and employ the fact that v is a solution to the PDE, which gives

$$\begin{aligned} \frac{d}{dt} \frac{1}{2} \|v_-\|_{L^2(\Omega)}^2 &= - \int_{\Omega} v_t v_- = D_v \int_{\Omega} \nabla v \cdot \nabla v_- + \delta_v \int_{\Omega} v v_- - \int_{\Gamma} (P_v - k_{\text{on}} u^2 v + k_{\text{off}} u_b) v_- \, dS \\ &= -D_v \|\nabla v_-\|_{L^2(\Omega; \mathbb{R}^d)}^2 - \delta_v \|v_-\|_{L^2(\Omega)}^2 - P_v \int_{\Gamma} v_- \, dS + k_{\text{on}} \int_{\Gamma} u^2 v v_- \, dS - k_{\text{off}} \int_{\Gamma} u_b v_- \, dS \\ &= -D_v \|\nabla v_-\|_{L^2(\Omega; \mathbb{R}^d)}^2 - \delta_v \|v_-\|_{L^2(\Omega)}^2 - P_v \int_{\Gamma} v_- \, dS - k_{\text{on}} \int_{\Gamma} u^2 (v_-)^2 \, dS - k_{\text{off}} \int_{\Gamma} u_b v_- \, dS \\ &\leq 0, \end{aligned}$$

where we used the non-negativity of v_- and u_b in the last step. By Theorem 2.18, we know that v is continuous in time with respect to the L^2 -norm. The only way for this to be achieved while also guaranteeing the time derivative is non-positive, is by having

$$\|v_-\|_{L^2(\Omega)} = 0$$

for almost every t , and thus v being non-negative.

To show that the map $t \mapsto \|v(t)\|_{L^2(\Omega)}^2$ is essentially bounded, we use v itself as a test function to get

$$\begin{aligned} \frac{d}{dt} \frac{1}{2} \|v\|_{L^2(\Omega)}^2 &= \int_{\Omega} v_t v = \int_{\Gamma} (P_v - k_{\text{on}} u^2 v + k_{\text{off}} u_b) v \, dS - D_v \int_{\Omega} \nabla v \cdot \nabla v - \delta_v \int_{\Omega} v v \\ &= P_v \int_{\Gamma} v \, dS - k_{\text{on}} \int_{\Gamma} u^2 v^2 \, dS + k_{\text{off}} \int_{\Gamma} u_b v \, dS - D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2 - \delta_v \|v\|_{L^2(\Omega)}^2 \\ &\leq P_v \int_{\Gamma} v \, dS + k_{\text{off}} \int_{\Gamma} u_b v \, dS - D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2. \end{aligned}$$

Now we apply Young's equality with ε to the first two integrals in this last expression to derive

$$\begin{aligned} P_v \int_{\Gamma} v \, dS &\leq \int_{\Gamma} \left(\varepsilon v^2 + \frac{1}{4\varepsilon} P_v^2 \right) \, dS = \varepsilon \|v\|_{L^2(\Gamma)}^2 + \frac{1}{4\varepsilon} P_v^2 |\Gamma| =: \varepsilon \|v\|_{L^2(\Omega)}^2 + \frac{1}{2} B_{\varepsilon}, \\ k_{\text{off}} \int_{\Gamma} u_b v &\leq \int_{\Gamma} \left(\varepsilon + \frac{1}{4\varepsilon} k_{\text{off}}^2 u_b^2 \right) \, dS = \varepsilon \|v\|_{L^2(\Gamma)}^2 + \frac{1}{4\varepsilon} k_{\text{off}}^2 \|u_b\|_{L^2(\Gamma)}^2 =: \varepsilon \|v\|_{L^2(\Gamma)}^2 + \frac{1}{2} D_{\varepsilon} \|u_b\|_{L^2(\Gamma)}^2, \end{aligned}$$

which, when plugged into the original estimate, gives

$$\frac{d}{dt} \frac{1}{2} \|v\|_{L^2(\Omega)}^2 \leq \frac{1}{2} B_\varepsilon + \frac{1}{2} D_\varepsilon \|w\|_{L^2(\Gamma)}^2 + 2\varepsilon \|v\|_{L^2(\Gamma)}^2 - D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2.$$

To convert the $L^2(\Gamma)$ -norm of v on the right-hand side into an $L^2(\Omega)$ -norm, we employ the trace inequality from Theorem 2.15 and get

$$\begin{aligned} \frac{d}{dt} \frac{1}{2} \|v\|_{L^2(\Omega)}^2 &\leq \frac{1}{2} B_\varepsilon + \frac{1}{2} D_\varepsilon \|w\|_{L^2(\Gamma)}^2 + 2\varepsilon C_\gamma^2 \|v\|_{H^1(\Omega)}^2 - D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2 \\ &= \frac{1}{2} B_\varepsilon + \frac{1}{2} D_\varepsilon \|w\|_{L^2(\Gamma)}^2 + 2\varepsilon C_\gamma^2 \|v\|_{L^2(\Omega)}^2 + 2\varepsilon C_\gamma^2 \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2 - D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2 \\ &\leq \frac{1}{2} B_\varepsilon + \frac{1}{2} D_\varepsilon \|w\|_{L^2(\Gamma)}^2 + 2\varepsilon C_\gamma^2 \|v\|_{L^2(\Omega)}^2 \\ &=: \frac{1}{2} B_\varepsilon + \frac{1}{2} D_\varepsilon \|w\|_{L^2(\Gamma)}^2 + \frac{1}{2} A_\varepsilon \|v\|_{L^2(\Omega)}^2 \end{aligned}$$

for small enough $\varepsilon > 0$ and almost every t . We are now ready to use Gronwall's lemma to get the final estimate

gronwall application

$$\begin{aligned} \|v(t)\|_{L^2(\Omega)}^2 &\leq \|v_0\|_{L^2(\Omega)}^2 e^{A_\varepsilon t} + \int_0^t \left(B_\varepsilon + D_\varepsilon \|w(s)\|_{L^2(\Gamma)}^2 \right) e^{A_\varepsilon(t-s)} ds \\ &= \|v_0\|_{L^2(\Omega)}^2 e^{A_\varepsilon t} + \frac{B_\varepsilon}{A_\varepsilon} (e^{A_\varepsilon t} - 1) + D_\varepsilon \int_0^t \|w(s)\|_{L^2(\Gamma)}^2 e^{A_\varepsilon(t-s)} ds \\ &\leq \|v_0\|_{L^2(\Omega)}^2 e^{A_\varepsilon t} + \frac{B_\varepsilon}{A_\varepsilon} (e^{A_\varepsilon t} - 1) + \frac{D_\varepsilon M_{u_b}(T)^2 |\Gamma|}{A_\varepsilon} (e^{A_\varepsilon t} - 1) \\ &\leq \|v_0\|_{L^2(\Omega)}^2 e^{A_\varepsilon T} + \frac{B_\varepsilon}{A_\varepsilon} (e^{A_\varepsilon T} - 1) + \frac{D_\varepsilon M_{u_b}(T)^2 |\Gamma|}{A_\varepsilon} (e^{A_\varepsilon T} - 1) \\ &=: M_v(T) \end{aligned}$$

for almost every t , where $M_{u_b}(T)$ was defined in the proof of Lemma 4.3. $\square$

Using Lemmas 4.3 and 4.4, we can define two operators between $\mathcal{Y}$ and $\mathcal{X}^2$. One takes two arbitrary functions $(u^*, u_b^*) \in \mathcal{X}^2$ and maps them to the unique solution $v \in \mathcal{Y}$ to Equation (4.8). Let us refer to this operator as $\mathcal{T}_1 : \mathcal{X}^2 \rightarrow \mathcal{Y}$. The other operator, $\mathcal{T}_2 : \mathcal{Y} \rightarrow \mathcal{X}^2$, takes a function $v^* \in \mathcal{Y}$ and defines $\mathcal{T}_2(v^*)$ as the unique solution $(u, u_b) \in \mathcal{X}^2$ of Equation (4.7). Combining these two operators, we define

$$\mathcal{T} := \mathcal{T}_2 \circ \mathcal{T}_1 : \mathcal{X}^2 \rightarrow \mathcal{X}^2.$$

We want to show that this operator has a unique fixed point. If this is the case, we have functions $(u, u_b) \in \mathcal{X}^2$ such that $\mathcal{T}(u, u_b) = (u, u_b)$, and let $v \in \mathcal{Y}$ denote $v = \mathcal{T}_1(u, u_b)$. Given that then $\mathcal{T}_2(v) = (u, u_b)$, by definition this means that v solves Equation (4.4) for almost every $t \in (0, T)$, and $w = (u, u_b)$ solves Equation (4.5) for almost every $t \in (0, T)$ and $x \in \Gamma$, meaning that v, u and u_b are weak solutions of our model. Conversely, if v, u and u_b are weak solutions to the model, then by the uniqueness of solutions in Lemmas 4.3 and 4.4, we must have that $\mathcal{T}_2(v) = (u, u_b)$ and $\mathcal{T}_1(u, u_b) = v$, and thus $\mathcal{T}(u, u_b) = (u, u_b)$.

It should be clear now how we will apply Banach's fixed-point theorem. If we prove that $\mathcal{T}$ is a contraction, we can guarantee the existence of a fixed point and therefore a weak solution. Since

every weak solution of the model is also a fixed point, the theorem also grants us uniqueness of such solutions. To prove that $\mathcal{T}$ is a contraction, we first show this for $\mathcal{T}_2$ and $\mathcal{T}_1$ separately, after which we can easily combine the results.

Lemma 4.5

For $T > 0$ small enough, $\mathcal{T}_1 : \mathcal{X}^2 \rightarrow \mathcal{Y}$ is a contraction. That is, there is a $q \in [0, 1)$ such that

$$\|\mathcal{T}_1(\bar{u}, \bar{u}_b) - \mathcal{T}_1(\tilde{u}, \tilde{u}_b)\|_{\mathcal{Y}}^2 \leq q \|(\bar{u}, \bar{u}_b) - (\tilde{u}, \tilde{u}_b)\|_{\mathcal{X}^2}^2$$

for every $(\bar{u}, \bar{u}_b), (\tilde{u}, \tilde{u}_b) \in \mathcal{X}^2$.

Proof. Let $(\bar{u}, \bar{u}_b), (\tilde{u}, \tilde{u}_b) \in \mathcal{X}^2$ and define $\bar{v}, \tilde{v} \in \mathcal{Y}$ by $\bar{v} := \mathcal{T}_1(\bar{u}, \bar{u}_b)$ and $\tilde{v} := \mathcal{T}_1(\tilde{u}, \tilde{u}_b)$. We now denote their differences by

$$v := \bar{v} - \tilde{v}, \quad u := \bar{u} - \tilde{u} \quad \text{and} \quad u_b := \bar{u}_b - \tilde{u}_b,$$

such that

$$\|\mathcal{T}_1(\bar{u}, \bar{u}_b) - \mathcal{T}_1(\tilde{u}, \tilde{u}_b)\|_{\mathcal{Y}}^2 = \|\bar{v} - \tilde{v}\|_{\mathcal{Y}}^2 = \|v\|_{\mathcal{Y}}^2.$$

We can estimate this norm using the weak form of the PDE. We use Theorem 2.18 to see that

stefanie comment

$$\frac{d}{dt} \frac{1}{2} \|v\|_{L^2(\Omega)}^2 = \int_{\Omega} v_t v = -D_v \int_{\Omega} |\nabla v|^2 - \delta_v \int_{\Omega} v^2 + \int_{\Gamma} (-k_{\text{on}} \bar{u}^2 \bar{v} + k_{\text{off}} \bar{u}_b + k_{\text{on}} \tilde{u}^2 \tilde{v} - k_{\text{off}} \tilde{u}_b) v \, dS,$$

which after rearranging becomes

$$\frac{d}{dt} \frac{1}{2} \|v\|_{L^2(\Omega)}^2 + D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2 + \delta_v \|v\|_{L^2(\Omega)}^2 = -k_{\text{on}} \int_{\Gamma} (\bar{u}^2 \bar{v} - \tilde{u}^2 \tilde{v}) v \, dS + k_{\text{off}} \int_{\Gamma} u_b v \, dS$$

for almost all t . We then use the identity

$$\bar{u}^2 \bar{v} - \tilde{u}^2 \tilde{v} = \bar{u}^2 \bar{v} - \tilde{u}^2 \bar{v} + \tilde{u}^2 \bar{v} - \tilde{u}^2 \tilde{v} = (\bar{u}^2 - \tilde{u}^2) \bar{v} + \tilde{u}^2 v = (\bar{u} + \tilde{u}) \bar{v} u + \tilde{u}^2 v \quad (4.9)$$

to estimate

$$\begin{aligned} & \frac{d}{dt} \frac{1}{2} \|v\|_{L^2(\Omega)}^2 + D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2 + \delta_v \|v\|_{L^2(\Omega)}^2 \\ &= -k_{\text{on}} \int_{\Gamma} (\bar{u} + \tilde{u}) \bar{v} u v \, dS - k_{\text{on}} \int_{\Gamma} \tilde{u}^2 v^2 \, dS + k_{\text{off}} \int_{\Gamma} u_b v \, dS \\ &\leq k_{\text{on}} \int_{\Gamma} |(\bar{u} + \tilde{u}) \bar{v} u v| \, dS + k_{\text{off}} \int_{\Gamma} u_b v \, dS \\ &\leq 2k_{\text{on}} M_u(T) M_v(T) \int_{\Gamma} |u v| \, dS + k_{\text{off}} \int_{\Gamma} u_b v \, dS. \end{aligned}$$

The mixed terms can be separated using Young's inequality, giving

$$\begin{aligned}
& \frac{d}{dt} \frac{1}{2} \|v\|_{L^2(\Omega)}^2 + D_v \|\nabla v\|_{L^2(\Omega; \mathbb{R}^d)}^2 + \delta_v \|v\|_{L^2(\Omega)}^2 \\
& \leq 2k_{\text{on}} M_u(T) M_v(T) \left(A_\varepsilon \|u\|_{L^2(\Gamma)}^2 + \varepsilon \|v\|_{L^2(\Gamma)}^2 \right) + k_{\text{off}} \left(A_\varepsilon \|u_b\|_{L^2(\Gamma)}^2 + \varepsilon \|v\|_{L^2(\Gamma)}^2 \right) \\
& \leq B_\varepsilon(T) \|u\|_{L^2(\Gamma)}^2 + k_{\text{off}} A_\varepsilon \|u_b\|_{L^2(\Gamma)}^2 + C(T) \varepsilon \|v\|_{L^2(\Gamma)}^2 \\
& \leq B_\varepsilon(T) \|u\|_{L^2(\Gamma)}^2 + k_{\text{off}} A_\varepsilon \|u_b\|_{L^2(\Gamma)}^2 + C_\gamma^2 C(T) \varepsilon \|v\|_{H^1(\Omega)}^2
\end{aligned}$$

for almost all t . Now let $\delta := \min(\delta_v, D_v)$ and integrate from 0 to T to get

$$\frac{1}{2} \|v(t)\|_{L^2(\Omega)}^2 + (\delta - C_\gamma^2 C(T) \varepsilon) \|v\|_{L^2(0,T;H^1(\Omega))}^2 \leq B_\varepsilon(T) \|u\|_{L^2(0,T;L^2(\Gamma))}^2 + k_{\text{off}} A_\varepsilon \|u_b\|_{L^2(0,T;L^2(\Gamma))}^2$$

for almost all t . If we now choose $\varepsilon < \frac{\delta}{C_\gamma^2 C(T)}$, we can estimate both the $L^2(0, T; H^1(\Omega))$ -norm and the $L^2(0, T; L^2(\Omega))$ -norm of v . The first will only be helpful later, but the latter concludes this proof. First, since $\delta - C_\gamma^2 C(T) \varepsilon > 0$, we have

$$\|v\|_{L^2(0,T;H^1(\Omega))}^2 \leq D_\varepsilon(T) \left(\|u\|_{L^2(0,T;L^2(\Gamma))}^2 + \|u_b\|_{L^2(0,T;L^2(\Gamma))}^2 \right). \quad (4.10)$$

Second, we can integrate from 0 to T once more to get

$$\|v\|_{L^2(0,T;L^2(\Omega))}^2 \leq T M_\varepsilon(T) \left(\|u\|_{L^2(0,T;L^2(\Gamma))}^2 + \|u_b\|_{L^2(0,T;L^2(\Gamma))}^2 \right).$$

If we trace back the definition of this constant $M_\varepsilon(T)$ in the proof of Lemma 4.3, we see that it is continuous in T with a finite value at $T = 0$. This means that for small enough T , we will have that $T M_\varepsilon(T) < 1$, finishing this proof. $\square$

We don't
know $M_v(T)$

Lemma 4.6

For $T > 0$ small enough, $\mathcal{T}_2 : \mathcal{Y} \rightarrow \mathcal{X}^2$ is a contraction. That is, there is a $q \in [0, 1)$ such that

$$\|\mathcal{T}_2(\bar{v}) - \mathcal{T}_2(\tilde{v})\|_{\mathcal{X}^2}^2 \leq q \|\bar{v} - \tilde{v}\|_{\mathcal{Y}}^2$$

for every $\bar{v}, \tilde{v} \in \mathcal{Y}$.

Proof. Now let $\bar{v}, \tilde{v} \in \mathcal{Y}$ and define $\bar{u}, \bar{u}_b, \tilde{u}$ and $\tilde{u}_b$ by $(\bar{u}, \bar{u}_b) := \mathcal{T}_2(\bar{v})$ and $(\tilde{u}, \tilde{u}_b) := \mathcal{T}_2(\tilde{v})$. Once again, we denote their differences by

$$v := \bar{v} - \tilde{v}, \quad u := \bar{u} - \tilde{u} \quad \text{and} \quad u_b := \bar{u}_b - \tilde{u}_b,$$

such that

$$\|\mathcal{T}_2(\bar{v}) - \mathcal{T}_2(\tilde{v})\|_{\mathcal{X}^2}^2 = \|(\bar{u}, \bar{u}_b) - (\tilde{u}, \tilde{u}_b)\|_{\mathcal{X}^2}^2 = \|(u, u_b)\|_{\mathcal{X}^2}^2 = \|u\|_{\mathcal{X}}^2 + \|u_b\|_{\mathcal{X}}^2.$$

We therefore seek to estimate these separate differences u and u_b in terms of v . Starting off with u , we see that it satisfies

first step?

$$\begin{aligned}
\frac{d}{dt} \|u\|_{L^2(\Gamma)}^2 &= 2 \int_{\Gamma} u_t u \, dS \\
&= 2 \int_{\Gamma} (\bar{u}_t - \tilde{u}_t) u \, dS \\
&= 2 \int_{\Gamma} (\alpha(\bar{u}_b - \tilde{u}_b) - \delta_u(\bar{u} - \tilde{u}) - 2k_{\text{on}}(\bar{u}^2 \bar{v} - \tilde{u}^2 \tilde{v}) + 2k_{\text{off}}(\bar{u}_b - \tilde{u}_b)) u \, dS \\
&= 2(\alpha + 2k_{\text{off}}) \int_{\Gamma} u u_b \, dS - 2\delta_u \int_{\Gamma} u^2 \, dS - 4k_{\text{on}} \int_{\Gamma} (\bar{u}^2 \bar{v} - \tilde{u}^2 \tilde{v}) u \, dS
\end{aligned}$$

for almost every t . As we did for the previous lemma, we now use the identity from Equation (4.9) to further derive

$$\begin{aligned}
\frac{d}{dt} \|u\|_{L^2(\Gamma)}^2 &= 2(\alpha + 2k_{\text{off}}) \int_{\Gamma} u u_b \, dS - 2\delta_u \int_{\Gamma} u^2 \, dS - \overbrace{4k_{\text{on}} \int_{\Gamma} (\bar{u} + \tilde{u}) \bar{v} u^2 \, dS}^{\geq 0} - 4k_{\text{on}} \int_{\Gamma} \tilde{u}^2 v u \, dS \\
&\leq 2(\alpha + 2k_{\text{off}}) \int_{\Gamma} u u_b \, dS - 2\delta_u \int_{\Gamma} u^2 \, dS + 4k_{\text{on}} \int_{\Gamma} \tilde{u}^2 |v u| \, dS \\
&\leq 2(\alpha + 2k_{\text{off}}) \int_{\Gamma} u u_b \, dS - 2\delta_u \|u\|_{L^2(\Gamma)}^2 + 4k_{\text{on}} M_u(T)^2 \int_{\Gamma} |v u| \, dS.
\end{aligned}$$

To further separate the mixed terms, we use Young's inequality twice and estimate

$$\begin{aligned}
\frac{d}{dt} \|u\|_{L^2(\Gamma)}^2 &\leq A_1(T) \|u\|_{L^2(\Gamma)}^2 + A_2 \|u_b\|_{L^2(\Gamma)}^2 + \widetilde{B}_1(T) \|v\|_{L^2(\Gamma)}^2, \\
&\leq A_1(T) \|u\|_{L^2(\Gamma)}^2 + A_2 \|u_b\|_{L^2(\Gamma)}^2 + B_1(T) \|v\|_{H^1(\Omega)}^2,
\end{aligned}$$

where $A_1(T)$, A_2 , $\widetilde{B}_1(T)$ and $B_1(T)$ are defined as

$$\begin{aligned}
A_1(T) &:= -2\delta_u + \alpha + 2k_{\text{off}} + 2k_{\text{on}} M_u(T)^2, & A_2 &:= \alpha + 2k_{\text{off}} \\
\widetilde{B}_1(T) &:= 2k_{\text{on}} M_u(T)^2 & \text{and} & B_1(T) := C_{\gamma}^2 \widetilde{B}_1(T).
\end{aligned}$$

Using the same arguments, we get a similar estimate for $\frac{d}{dt} \|u_b\|_{L^2(\Gamma)}^2$, namely

$$\begin{aligned}
\frac{d}{dt} \|u_b\|_{L^2(\Gamma)}^2 &= 2 \int_{\Gamma} ((\bar{u}_b)_t - (\tilde{u}_b)_t) u_b \, dS \\
&= -2(\delta_{u_b} + k_{\text{off}}) \int_{\Gamma} u_b^2 \, dS + 2k_{\text{on}} \int_{\Gamma} (\bar{u}^2 \bar{v} - \tilde{u}^2 \tilde{v}) u_b \, dS \\
&= -2(\delta_{u_b} + k_{\text{off}}) \int_{\Gamma} u_b^2 \, dS + 2k_{\text{on}} \int_{\Gamma} (\bar{u} + \tilde{u}) \tilde{v} u u_b \, dS + 2k_{\text{on}} \int_{\Gamma} \tilde{u}^2 v u_b \, dS \\
&\leq -2(\delta_{u_b} + k_{\text{off}}) \int_{\Gamma} u_b^2 \, dS + 2k_{\text{on}} \int_{\Gamma} (\bar{u} + \tilde{u}) \tilde{v} |u u_b| \, dS + 2k_{\text{on}} \int_{\Gamma} \tilde{u}^2 |v u_b| \, dS \\
&\leq -2(\delta_{u_b} + k_{\text{off}}) \int_{\Gamma} u_b^2 \, dS + 4k_{\text{on}} M_u(T) M_v(T) \int_{\Gamma} |u u_b| \, dS + 2k_{\text{on}} M_u(T)^2 \int_{\Gamma} |v u_b| \, dS
\end{aligned}$$

$$\begin{aligned}
&\leq A_3(T) \|u_b\|_{L^2(\Gamma)}^2 + A_4(T) \|u\|_{L^2(\Gamma)}^2 + \widetilde{B}_2(T) \|v\|_{L^2(\Gamma)}^2, \\
&\leq A_3(T) \|u_b\|_{L^2(\Gamma)}^2 + A_4(T) \|u\|_{L^2(\Gamma)}^2 + B_2(T) \|v\|_{H^1(\Omega)}^2,
\end{aligned}$$

where $A_3(T)$, $A_4(T)$, $\widetilde{B}_2(T)$ and $B_2(T)$ are given by

$$\begin{aligned}
A_3(T) &:= 2k_{\text{on}} M_u(T) M_v(T) + k_{\text{on}} M_u(T)^2 - 2\delta_{u_b} - 2k_{\text{off}}, \\
A_4(T) &:= 2k_{\text{on}} M_u(T) M_v(T), \quad \widetilde{B}_2(T) := k_{\text{on}} M_u(T)^2 \quad \text{and} \quad B_2(T) := C_\gamma^2 \widetilde{B}_2(T).
\end{aligned}$$

To merge these estimates, we let $A(T) := 2 \max(A_1(T), A_2, A_3(T), A_4(T))$ and $B(T) := 2 \max(B_1(T), B_2(T))$ and sum to get overflow

$$\frac{d}{dt} \left(\|u\|_{L^2(\Gamma)}^2 + \|u_b\|_{L^2(\Gamma)}^2 \right) \leq A(T) \left(\|u\|_{L^2(\Gamma)}^2 + \|u_b\|_{L^2(\Gamma)}^2 \right) + B(T) \|v\|_{H^1(\Omega)}^2$$

for almost all t . An application of Gronwall's lemma gives gronwall application

$$\begin{aligned}
\|u(t)\|_{L^2(\Gamma)}^2 + \|u_b(t)\|_{L^2(\Gamma)}^2 &\leq \|u(0)\|_{L^2(\Gamma)}^2 + \|u_b(0)\|_{L^2(\Gamma)}^2 + B(T) \int_0^t \|v(s)\|_{H^1(\Omega)}^2 e^{(t-s)A(T)} ds \\
&\leq B(T) e^{TA(T)} \|v\|_{L^2(0,T;H^1(\Omega))}^2,
\end{aligned}$$

since the initial conditions of $\bar{u}$ and $\tilde{u}$ are the same, and therefore $u(0) = 0$ for every $x \in \Gamma$. Naturally, the same holds for $u_b(0)$. After integrating from 0 to T , we can conclude that

$$\|u\|_{L^2(0,T;L^2(\Gamma))}^2 + \|u_b\|_{L^2(0,T;L^2(\Gamma))}^2 \leq TB(T) e^{TA(T)} \|v\|_{L^2(0,T;H^1(\Omega))}^2.$$

It may seem like we are done now, but the current estimate uses the wrong norm on v . To get an estimate using the $L^2(0, T; L^2(\Omega))$ -norm instead, we can use Equation (4.10) to get

$$\begin{aligned}
&\|u\|_{L^2(0,T;L^2(\Gamma))}^2 + \|u_b\|_{L^2(0,T;L^2(\Gamma))}^2 \\
&\leq TB(T) e^{TA(T)} D_\varepsilon(T) \left(\|u\|_{L^2(0,T;L^2(\Gamma))}^2 + \|u_b\|_{L^2(0,T;L^2(\Gamma))}^2 \right) + \overbrace{TB(T) e^{TA(T)} \|v\|_{L^2(0,T;L^2(\Omega))}^2}^{\geq 0},
\end{aligned} \tag{4.11}$$

or after rearranging terms,

$$\|u\|_{L^2(0,T;L^2(\Gamma))}^2 + \|u_b\|_{L^2(0,T;L^2(\Gamma))}^2 \leq \frac{TB(T) e^{TA(T)}}{1 - TB(T) e^{TA(T)} D_\varepsilon(T)} \|v\|_{L^2(0,T;L^2(\Omega))}^2.$$

As this factor clearly approaches 0 when T goes to 0, we can choose T sufficiently small such that it is less than 1 and conclude the proof. □

We have now gathered enough results to prove the main statement of this section.

Proof of Theorem 4.2. Let $\tilde{T} > 0$ be small enough such that, by Lemmas 4.5 and 4.6, $\mathcal{T}_1$ and $\mathcal{T}_2$ are contractions with Lipschitz constants q_1 and q_2 , respectively. Define, as discussed before, $\mathcal{T}$ as the composition of $\mathcal{T}_1$ and $\mathcal{T}_2$. Then for any $(\bar{u}, \bar{u}_b), (\tilde{u}, \tilde{u}_b) \in \mathcal{X}^2$, we have

$$\|\mathcal{T}(\bar{u}, \bar{u}_b) - \mathcal{T}(\tilde{u}, \tilde{u}_b)\|_{\mathcal{X}^2}^2 \leq q_2 \|\mathcal{T}_1(\bar{u}, \bar{u}_b) - \mathcal{T}_1(\tilde{u}, \tilde{u}_b)\|_{\mathcal{Y}}^2 \leq q_2 q_1 \|(\bar{u}, \bar{u}_b) - (\tilde{u}, \tilde{u}_b)\|_{\mathcal{X}^2}^2,$$

showing that $\mathcal{T}$ is a contraction since $q_2 q_1 \in [0, 1)$.

We apply Banach's fixed-point theorem to get a unique fixed point $(u, u_b) \in \mathcal{X}^2$. As previously discussed, this corresponds to a unique weak solution on $[0, \tilde{T}]$. By setting new initial conditions $u_0 = u(T)$, $u_{b,0} = u_b(T)$ and $v_0 = v(T) := \mathcal{T}_1(u, u_b)(T)$ and applying the same process, this solution can be extended to $[0, \bar{T}]$ for some $\bar{T} > \tilde{T}$. We continue repeating these steps until a maximum interval $I_{\max}$ is reached, at which point we distinguish three cases.

First of all, if $[0, T] \subseteq I_{\max}$, we are immediately done, as we now have a solution on $[0, T]$ with the required properties. If $I_{\max} = [0, T_{\max}]$ for some $T_{\max} < T$, we can set new initial conditions at $u(T_{\max})$, $u_b(T_{\max})$ and $v(T_{\max})$ and extend again, contradicting the fact that $I_{\max}$ is the maximum interval of existence. Lastly, if instead $I_{\max} = [0, T_{\max})$ for some $T_{\max} < T$, we apply Theorem 2.18 to extend v to a continuous function on $[0, T_{\max}]$. This allows us to use the same method again and extend the solution. □

flip order

Chapter 5

Numerical analysis

We have seen several qualitative properties of solutions to our model. Some properties, however, are more difficult to investigate using purely analytical methods. One of these properties is the ability to generate Turing patterns. We therefore turn to numerical methods and simulate the solutions of the model to have a more concrete and clear view of its behaviour.

5.1 Modified model

5.1.1 Diffusion on the surface

Now that we are working with simulations, we can make our model slightly more complex by dropping one of our assumptions. Let Ω and Γ be constructed as in Section 4.1. In this chapter, we restrict our study to the dimensions of $d = 2$ and $d = 3$, since these are the most relevant to the biological applications of this model. Recall that in Section 4.1, we assumed the receptors, both free and bounded, to be fixed in place on the cell membrane. This resulted in a system of one PDE and two ODEs, the latter being both defined separately for every point on the cell membrane. We now drop this assumption and add diffusion of free and bounded receptors, leading to a new system of reaction diffusion PDEs given by

$$\begin{aligned}
 u_t &= D_u \Delta u + \gamma (P_u + \alpha u_b - \delta_u u - 2k_{\text{on}} u^2 v + 2k_{\text{off}} u_b) && \text{on } (0, T) \times \Gamma, \\
 u_{b,t} &= D_{u_b} \Delta u_b + \kappa \gamma (-\delta_{u_b} u_b + k_{\text{on}} u^2 v - k_{\text{off}} u_b) && \text{on } (0, T) \times \Gamma, \\
 v_t &= D_v \Delta v - \gamma \delta_v v && \text{in } (0, T) \times \Omega, \\
 D_v \frac{\partial v}{\partial \nu} &= \gamma (P_v - k_{\text{on}} u^2 v + k_{\text{off}} u_b) && \text{on } (0, T) \times \Gamma, \\
 D_v \frac{\partial v}{\partial \nu} &= 0 && \text{on } (0, T) \times \Gamma_e,
 \end{aligned} \tag{5.1}$$

where D_u and D_{u_b} are positive. We have also added two reaction scaling coefficients $\gamma > 0$ and $\kappa > 0$. When Turing instabilities can take place in the system, we will adjust γ such that the domain is sufficiently large and the patterns become visible. The second scaling coefficient κ is used to simulate the quasi-steady state assumption we will discuss in the next subsection. The corresponding initial conditions are

$$u(0, x) = u_0(x), \quad u_b(0, x) = u_{b,0}(x) \quad \text{and} \quad v(0, y) = v_0(y)$$

for all $x \in \Gamma$ and $y \in \bar{\Omega}$.

There are multiple things to note about Equation (5.1). Since Γ has an empty interior, it is impossible to define spatial derivatives for functions living exclusively on this set. This means we would not be able to use the Laplace operator to include diffusion of the receptors as we do in Equation (5.1). This issue is solved by using the so called the Laplace-Beltrami operator instead, which is a generalisation of the Laplace operator to Riemannian manifolds. Since we assumed Γ to be a C^1 boundary, it forms such a manifold. The Laplace-Beltrami operator is further discussed in [Jos17], but we will not go into details here. For our purposes, it is sufficient to note that it behaves similarly to the classical Laplace operator and admits analogous rules of calculus such as partial integration. It is sometimes denoted by Δ_Γ , but for the sake of simplicity, we will omit this notation and use Δ for both the Laplace and Laplace-Beltrami operators, but they are in fact different operators.

Since we now have PDEs for u and u_b , it seems necessary to impose certain boundary conditions. However, recalling our exact construction of the domain, we can see that the set Γ representing the cell membranes, when viewed as a separate $(d - 1)$ -dimensional manifold, does not have a boundary. Since the equations for u and u_b strictly live on this manifold, boundary conditions are not required.

5.1.2 Derivation of the two-species model

We seek to compare this three-species bulk-surface model to a two-species model similar to that derived in [KMI14]. To obtain such a two-species model, we make some of the same assumptions. The first and most impactful is that the dynamics of the bounded receptors are so fast that $u_{b,t}$ and Δu_b are negligible. In the simulations, this will be realised by setting $\kappa \gg 1$. This results in an explicit expression for u_b in terms of u and v ,

$$u_b = \frac{k_{\text{on}}}{\delta_{u_b} + k_{\text{off}}} u^2 v.$$

This is called a *quasi-steady state*. This expression enables us to write a two-species model

$$\begin{aligned} u_t &= D_u \Delta u + \gamma (P_u - \delta_u u + \beta u^2 v) && \text{on } (0, T) \times \Gamma, \\ v_t &= D_v \Delta v - \gamma \delta_v v && \text{in } (0, T) \times \Omega, \\ D_v \frac{\partial v}{\partial \nu} &= \gamma (P_v - \beta u^2 v) && \text{on } (0, T) \times \Gamma, \\ D_v \frac{\partial v}{\partial \nu} &= 0 && \text{on } (0, T) \times \Gamma_e, \end{aligned} \tag{5.2}$$

where $\beta := \frac{k_{\text{on}} \delta_{u_b}}{\delta_{u_b} + k_{\text{off}}}$ and we make the second assumption, namely that $\alpha = 3\beta$. The final assumption made in [KMI14] is that $\delta_v v \ll \beta u^2 v$, making the natural decay of the ligands negligible and recovering the classical Schnakenberg equations. However, as we saw in Section 3.3, the Schnakenberg equations with additional decay of v is also able to generate patterns, so we will not be making this assumption.

5.2 The Finite Element Method

Numerical methods for ODEs typically include defining a small time step and using the ODE to derive a suitable iteration. In the study of PDEs, one might think to extend this approach to the spacial variable. This would result in defining a time step $\tau > 0$ as well as a space discretisation $h > 0$ and discretising the domain $(0, T) \times \Omega$ as a subset of $\tau\mathbb{Z} \times h\mathbb{Z}^d$. One method that uses this approach is the finite *difference* method, which is very successfully applied to a wide range of PDEs. However, this method is fairly hard to use when the dimension gets higher than $d = 1$ or $d = 2$, or the shape of the domain becomes more complex than, for instance, a rectangle. Our model is defined on a domain that contains holes that are designed to model cells and as a result, will contain some curvature. The finite difference method is too unreliable in this case, so we need a more robust and adaptable approach.

5.2.1 Basic construction

The finite *element* method is another way to approximate solutions to a PDE. It is mainly used for discretising the space variable, so let us focus on an example that does not include time. We consider the standard Poisson equation

$$\Delta u = f \quad \text{in } \Omega \quad (5.3)$$

with Neumann boundary conditions

$$\frac{\partial u}{\partial \nu} = g \quad \text{on } \partial\Omega. \quad (5.4)$$

Here, $\Omega \subseteq \mathbb{R}^d$ is an open and bounded set with C^1 boundary and f and g are known real-valued functions on Ω and $\partial\Omega$, respectively. We first derive the weak form of this PDE. Multiply Equation (5.3) with a smooth function $\phi \in C^\infty(\Omega)$ and integrate to get

$$\int_{\Omega} f\phi = \int_{\Omega} \Delta u\phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \phi \, dS - \int_{\Omega} \nabla u \cdot \nabla \phi = \int_{\partial\Omega} g\phi \, dS - \int_{\Omega} \nabla u \cdot \nabla \phi.$$

A weak solution of Equations (5.3) and (5.4) then is a function $u \in H^1(\Omega)$ such that

$$\int_{\Omega} \nabla u \cdot \nabla \phi = \int_{\partial\Omega} g\phi \, dS - \int_{\Omega} f\phi \quad (5.5)$$

for every $\phi \in H^1(\Omega)$.

By discretising the spatial domain, we will transform Equation (5.5) into a system of linear equations. To this end, we use the same approach as in [DE13] and define a *mesh*. We choose a finite number of points $\mathcal{N}_h = \{x_i\}_{i=1}^m$ in $\overline{\Omega}$ and use them as vertices to define d -simplices¹, which we collect in a set $\mathcal{T}_h$. These points are called *nodes*, and the d -simplices they enclose are referred to as *elements*. To have a well-behaved discretisation, for any two distinct d -simplices $T, T' \in \mathcal{T}_h$ we require $T \cap T'$ to either be empty or form a common subsimplex of dimension $k < d$.

¹An n -simplex is the n -dimensional equivalent of a triangle. The most important instances are the 0-simplex, which is a point, the 1-simplex, a line segment, the 2-simplex, a triangle, and the 3-simplex, a tetrahedron. Every n -simplex is enclosed by $n + 1$ $(n - 1)$ -simplices, which in turn are enclosed by $(n - 2)$ -simplices, etc. All of these are referred to as its subsimplices.

This leaves us with an approximation of Ω , denoted by

$$\Omega_h := \bigcup_{T \in \mathcal{T}_h} T.$$

The subscript h derives from the idea that this still represents the fineness of our discretisation, just as in the finite difference method. If we define

$$h := \max_{T \in \mathcal{T}_h} \text{diam}(T),$$

we want Ω_h to approximate Ω increasingly well as h approaches 0.

example

Additionally, we want a discretised function space that we use to construct approximate solutions to Equations (5.3) and (5.4), called the *trial space*. To this end, for every node $x_i \in \mathcal{N}_h$, we define a basis function $\phi_i : \Omega_h \rightarrow \mathbb{R}$ such that $\phi_i(x_j) = \delta_{ij}$ for all $j \in \{1, \dots, m\}$ and ϕ_i is linearly interpolated on each element. Now the function space we consider for our trial functions is the space of linear combinations of these basis functions,

example

$$\mathcal{V}_{\Omega_h} := \text{span}\{\phi_1, \dots, \phi_m\},$$

which is a finite-dimensional subspace of $H^1(\Omega)$. This is called the space of *first-order Lagrange elements*, derived from the fact that the functions are piecewise linear polynomials on each element. There are many other kinds of function spaces used in finite element simulations, such as second-order Lagrange spaces, where the functions are piecewise quadratic polynomials on each element, or spaces where the functions are allowed to be discontinuous, referred to as discontinuous Galerkin elements.

$H^1(\Omega_h)$ better?

We can now use this setup to convert our original problem into one that a computer can handle. Let $u_h \in \mathcal{V}_{\Omega_h}$ be an approximate solution to Equation (5.5). That is, let u_h solve

reference

$$\int_{\Omega_h} \nabla u_h \cdot \nabla \phi = \int_{\partial\Omega_h} g \phi \, dS - \int_{\Omega_h} f \phi$$

for every $\phi \in \mathcal{V}_{\Omega_h}$ ². Then for any $i \in \{1, \dots, m\}$, we can set $\phi = \phi_i$ and write u_h as its linear decomposition with coefficients $U_1, \dots, U_m \in \mathbb{R}$, deriving

$$\int_{\partial\Omega_h} g \phi_i \, dS - \int_{\Omega_h} f \phi_i = \int_{\Omega_h} \nabla \left(\sum_{j=1}^m U_j \phi_j \right) \cdot \nabla \phi_i = \sum_{j=1}^m U_j \int_{\Omega_h} \nabla \phi_j \cdot \nabla \phi_i.$$

If we define

$$F_i := \int_{\partial\Omega_h} g \phi_i \, dS - \int_{\Omega_h} f \phi_i \quad \text{and} \quad K_{ij} := \int_{\Omega_h} \nabla \phi_j \cdot \nabla \phi_i$$

for every $i, j \in \{1, \dots, m\}$, this is equivalent to the linear system

$$KU = F. \tag{5.6}$$

Here, K is typically referred to as the *stiffness matrix*. We have now transformed the PDE into a linear system of equations, which we can solve using a wide variety of algorithms.

²Note that for this to be well-defined, we should also define some approximation of g and f such that they are properly defined on $\partial\Omega_h$ and Ω_h , respectively. We will not go into detail about that here.

5.2.2 Time-dependent PDEs

We should quickly illustrate how time discretisation ties into this approach. Consider, therefore, the heat equation

$$u_t = \Delta u + f \quad \text{in } (0, T) \times \Omega \quad (5.7)$$

with Neumann boundary conditions

$$\frac{\partial u}{\partial \nu} = g \quad \text{on } [0, T) \times \partial\Omega, \quad (5.8)$$

where $T > 0$ is some final time and now f and g are real-valued functions on $(0, T) \times \Omega$ and $[0, T) \times \partial\Omega$, respectively. Of course, we also need initial conditions

$$u(0, x) = u_0(x), \quad x \in \Omega. \quad (5.9)$$

From Section 2.5, we know that we can define a weak solution of this PDE as a function $u \in H^1(0, T; H^1(\Omega))$ such that Equation (2.6) holds for every $\phi \in H^1(\Omega)$ and almost every $t \in [0, T]$.

Similarly to the numerical study of ODEs, we now choose a number of steps $N \in \mathbb{N}$ and let $\tau := T/N$. For every $n \in \{1, \dots, N\}$, we seek to approximate the time step

$$u_n(x) := u(n\tau, x), \quad x \in \Omega.$$

To this end, we use a difference quotient to approximate the time derivative and write

$$\int_{\Omega} \frac{u_{n+1} - u_n}{\tau} \phi + \int_{\Omega} \nabla u_{n+1} \cdot \nabla \phi = \int_{\partial\Omega} g(n\tau, x) \phi(x) \, dS(x) + \int_{\Omega} f(n\tau, x) \phi(x) \, dx.$$

where we have used implicit diffusion and explicit forcing. This is a widely used scheme, commonly referred to as IMEX [ARS97, PR01, BPR25]. Other schemes can be used with varying results in numerical stability and accuracy, depending on what class of PDE it is applied to. In this thesis, we will not go too far into this choice.

After rearranging terms, this leads to the time-independent problem

$$\int_{\Omega} u_{n+1} \phi + \tau \int_{\Omega} \nabla u_{n+1} \cdot \nabla \phi = \tau \int_{\Gamma} g(n\tau, x) \phi(x) \, dS(x) + \int_{\Omega} u_n \phi + \tau \int_{\Omega} f(n\tau, x) \phi(x) \, dx \quad (5.10)$$

for $n = 0, \dots, N-1$. After substituting u_{n+1} and u_n with their linear decompositions and setting $\phi = \phi_i \in \mathcal{V}_{\Omega_h}$ for some $i \in \{1, \dots, m\}$, this is transformed to the system of equations

$$(M + \tau K)U^{n+1} = MU^n + \tau F^n,$$

where $U^n \in \mathbb{R}^m$ contains the coefficients for the linear decomposition of u_n into $\phi_1, \dots, \phi_m$, and F^n and M are defined by

$$F_i^n := \int_{\partial\Omega_h} g(n\tau, x) \phi_i(x) \, dS(x) + \int_{\Omega_h} f(n\tau, x) \phi_i(x) \, dx \quad \text{and} \quad M_{ij} := \int_{\Omega_h} \phi_j \phi_i$$

for $i, j \in \{1, \dots, m\}$. M is often called the *mass matrix*. When calculating a new iteration U^{n+1} , we can assume the previous iterations are known, starting off with the initial conditions

$$U_i^0 := u_0(x_i), \quad i = 0, \dots, m.$$

In this example, we first discretised time, leading to a PDE in space which was then discretised with the finite element method. This order can also be swapped by assuming that the approximate solution u can be written as a linear combination of functions in $\mathcal{V}_{\Omega_h}$ with time-dependent coefficients

$$u(t, x) = \sum_{j=1}^m U_j(t) \phi_j(x).$$

Then the process of integrating over Ω and using integration by parts leads to a system of linear ODEs for $U(t) = (U_1(t), \dots, U_m(t))$, called the *semi-discrete* form of the PDE [Tho84]. This approach is more similar to the Galerkin method for proving existence and uniqueness of weak solutions to PDEs as demonstrated in [Eva10]. The reason why we instead choose to illustrate discretising time first is that, as we will discuss later, the libraries we use to simulate the PDEs require us to input the form of Equation (5.10), after which it will derive the linear system of equations for us. This makes it more important for us to explicitly think about discretising time instead of first discretising space.

5.2.3 Bulk-surface equations

In a bulk-surface setting, interacting functions are defined on different domains that depend on each other, so we should carefully think about how to discretise them. Let $\Gamma \subseteq \partial\Omega$. When simulating a bulk-surface equation, we first discretise the bulk and extract the surface mesh from the bulk mesh by defining the set of *boundary facets* (or *exterior facets*)

$$\mathcal{F}_h^e := \{S \mid S \text{ is a } (d-1)\text{-dimensional subsimplex of some } T \in \mathcal{T}_h \text{ and every vertex of } S \text{ lies on } \Gamma\},$$

after which we define

$$\Gamma_h := \bigcup_{S \in \mathcal{F}_h^e} S \subseteq \partial\Omega_h,$$

analogous to the definition of Ω_h . This also generates a subset of boundary nodes $\mathcal{N}_h^\Gamma := \mathcal{N}_h \cap \Gamma_h = \{y_1, \dots, y_l\}$, on which we can define another collection of basis functions $\psi_1, \dots, \psi_l$ such that $\psi_i(y_j) = \delta_{ij}$ for every $i, j \in \{1, \dots, l\}$ and every ψ_i is linearly interpolated in between nodes. This defines a finite-dimensional subspace of $H^1(\Gamma)$ given by

$$\mathcal{V}_{\Gamma_h} := \text{span}\{\psi_1, \dots, \psi_l\}.$$

In the next section, this function space is used to derive the time discretisations for our bulk surface models.

5.3 Implementation

We now derive the time discretisations of Equations (5.1) and (5.2). For both models, we use an IMEX scheme where we choose implicit discretisation for all linear terms and explicit discretisation for the non-linear reaction. For Equation (5.2), this results in the iteration

$$(1 + \tau\gamma\delta_u) \int_{\Gamma_h} u_{n+1} \psi \, dS + \tau D_u \int_{\Gamma_h} \nabla u_{n+1} \cdot \nabla \psi \, dS = \int_{\Gamma_h} u_n \psi \, dS + \tau\gamma \int_{\Gamma_h} [P_u + \beta u_n^2 v_n] \psi \, dS \quad (5.11)$$

for the surface PDE, where we now let Γ_h denote the submesh of $\partial\Omega_h$ that approximates Γ , and

$$(1 + \tau\gamma\delta_v) \int_{\Omega_h} v_{n+1} \phi + \tau D_v \int_{\Omega_h} \nabla v_{n+1} \cdot \nabla \phi = \int_{\Omega_h} v_n \phi \, dS - \tau\gamma \int_{\Gamma_h} [P_u + \beta u_n^2 v_n] \phi \, dS \quad (5.12)$$

for the bulk PDE, where $\psi \in \mathcal{V}_{\Gamma_h}$ and $\phi \in \mathcal{V}_{\Omega_h}$. With the same approach, Equation (5.1) admits the time discretisation

define what a submesh is or use other notation in previous section

$$\begin{aligned} (1 + \tau\gamma\delta_u) \int_{\Gamma_h} u_{n+1} \psi \, dS + \tau D_u \int_{\Gamma_h} \nabla u_{n+1} \psi \, dS - \tau\gamma(\alpha + 2k_{\text{off}}) \int_{\Gamma_h} u_{b,n+1} \psi \, dS \\ = \int_{\Gamma_h} u_n \psi \, dS + \tau\gamma \int_{\Gamma_h} [P_u - 2k_{\text{on}} u_n^2 v_n] \psi \, dS \end{aligned} \quad (5.13)$$

for the PDE for u ,

$$\begin{aligned} (1 + \tau\kappa\gamma\delta_{u_b}) \int_{\Gamma_h} u_{b,n+1} \chi \, dS + \tau D_{u_b} \int_{\Gamma_h} \nabla u_{b,n+1} \chi \, dS + \tau\kappa\gamma k_{\text{off}} \int_{\Gamma_h} u_{b,n+1} \chi \, dS \\ = \int_{\Gamma_h} u_{b,n} \chi \, dS + \tau\kappa\gamma k_{\text{on}} \int_{\Gamma_h} u_n^2 v_n \chi \, dS \end{aligned} \quad (5.14)$$

for the PDE for u_b , and finally for v ,

$$\begin{aligned} (1 + \tau\gamma\delta_v) \int_{\Omega_h} v_{n+1} \phi + \tau D_v \int_{\Omega_h} \nabla v_{n+1} \phi - \tau\gamma k_{\text{off}} \int_{\Gamma_h} u_{b,n+1} \phi \, dS \\ = \int_{\Omega_h} v_n \phi + \tau\gamma \int_{\Gamma_h} [P_v - k_{\text{on}} u_n^2 v_n] \phi \, dS, \end{aligned} \quad (5.15)$$

where $\psi, \chi \in \mathcal{V}_{\Gamma_h}$ and $\phi \in \mathcal{V}_{\Omega_h}$.

The initial conditions are set as random perturbations of the uniform steady states. That is, we define $u_0, u_{b,0} \in \mathcal{V}_{\Gamma_h}$ and $v_0 \in \mathcal{V}_{\Omega_h}$ by

specify exact uniform steady states

$$u_0 = \sum_{i=1}^l (U + a_i) \psi_i, \quad u_{b,0} = \sum_{i=1}^l (U_b + b_i) \psi_i \quad \text{and} \quad v_0 = \sum_{i=1}^m (V + c_i) \phi_i,$$

where U, U_b and V are the uniform steady states for u, u_b and v , respectively, and $a_1, \dots, a_l, b_1, \dots, b_l$ and $c_1, \dots, c_m$ are uniformly sampled from $[-0.1, 0.1]$.

To calculate these iterations, Equations (5.11) to (5.15) are given as input to a Python scripts using FEniCSx, along with a mesh generated by Gmsh. FEniCSx derives the system of linear equations, which is solved with PETSc. All scripts that were written and used for this thesis can be found on the public GitHub page https://github.com/bregtvd103/Bachelor_Thesis_Turing_patterns.

5.4 Results

I need to redo some simulations, so I will insert the actual figures and parameter values later. We first simulate solutions to Equations (5.1) and (5.2) for $d = 2$ with Ω being the open square $(-16, 16)^2$ without the four closed disks of radius 4 and centers $(\pm 8, \pm 8)$ and Γ being set to the boundaries of these disks. In , the numerical solution of Equation (5.2) is plotted at different times. We can clearly see Turing patterns appearing on the cell membranes, similar to the ones generated by the Schnakenberg equations in Sections 3.2 and 3.3. These now represent clusters of receptors, with the lower regions representing relative scarcity.

figure

We compare this to the numerical solutions of Equation (5.1). To simulate the quasi-steady state assumption, we set $\kappa = 50$. This necessitates increasing the number of steps N significantly to preserve numerical stability. If the model of two species proves to be a good representation of the three-species model, it would be preferable to use this model rather than the full three-species equivalent, since it is simpler to analyse both numerically and analytically. In , the numerical solution of Equation (5.1) is plotted for similar times. It admits the same general behaviour as the numerical solution of Equation (5.2), smoothing out the perturbations before forming Turing patterns of similar scale and shape. The patterns formed by u and u_b almost coincide, with the peaks of u_b reaching only a slightly higher. This can be justified by the extra scaling coefficient κ dominating the diffusion.

figure

For $d = 3$, the domain is defined as the open cube $(-16, 16)^3$ without the eight closed spheres of radius 4 and centers $(\pm 8, \pm 8, \pm 8)$, and Γ is defined as the boundary of these spheres. In , we see the two-species system form patterns on the cell membranes and their effect on the concentration of ligands in the immediate neighbourhood of the cell membranes. Again, this behaviour is very similar to the numerical solution of the three-species model, shown in . The peaks of u are also similar to that of u_b again, with the peaks of u_b being slightly higher than those of u .

figure

figure

For both $d = 2$ and $d = 3$ and the chosen parameter values, the model of Equation (5.2) seems to be a good representation for that of Equation (5.1). Moreover, when choosing the two-species model to further study the behaviour of these PDEs, the concentration of u_b can be assumed to be similar to that of u . These conclusions only hold for the chosen parameter values. To extend them to the general case, more tests should be done, since every choice of parameters for Equation (5.2) admits an infinite set of possible parameter values for Equation (5.1) such that the quasi-steady state reduction produces the same model.

need to find a better way to phrase this?

Chapter 6

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